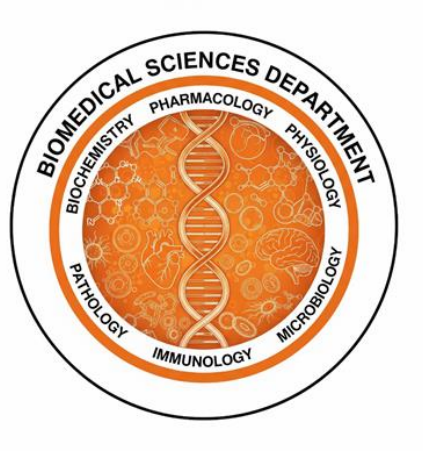
Nucleotide Metabolism & Gout I-II

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Fall 2026



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Objectives

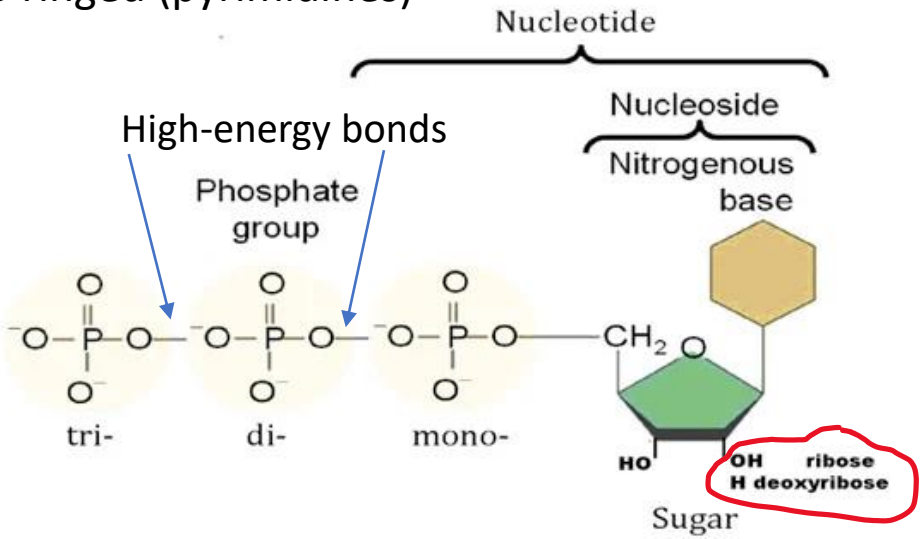
- **Review** the roles of nucleotides in living systems.
- **Compare and contrast** the structure, biosynthesis, salvage, and degradation of purines and pyrimidines.
- **Discuss** the role of folate in nucleotide metabolisms.
- **Explain** the mechanisms of drugs in relation to nucleotide metabolism in bacteria and humans.
- **Explain** the metabolic basis of disorders in nucleotide metabolism.
- **Identify** the mechanisms underlying nucleotide metabolism and gout.
- **Describe** gout diagnosis based on the optical properties of monosodium urate crystals.
- **Explain** the mechanisms of action of allopurinol.

Review: Roles of Nucleotides

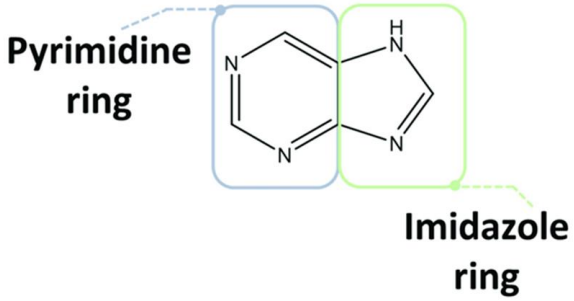
- **Monomeric units of nucleic acids (building blocks for DNA and RNA)**
  - **Carry the information needed for protein synthesis and cell proliferation**
- **Energy**: ATP (a purine nucleotide)
- **Carriers** of activated intermediates (i.e., UDP-glucose, CDP-choline)
- **Precursor** function: GTP - protects the 5' end of messenger RNA (mRNA) by capping it
- **Components of coenzymes**: NAD, FAD, FMN, coenzyme A
- **Allosteric effectors** of enzymes (inhibitors/activators)
- **Physiological mediators**: adenosine (coronary blood flow); ADP (platelet aggregation); cAMP and cGMP (secondary messengers); signal transduction via G-coupled receptors

Structures

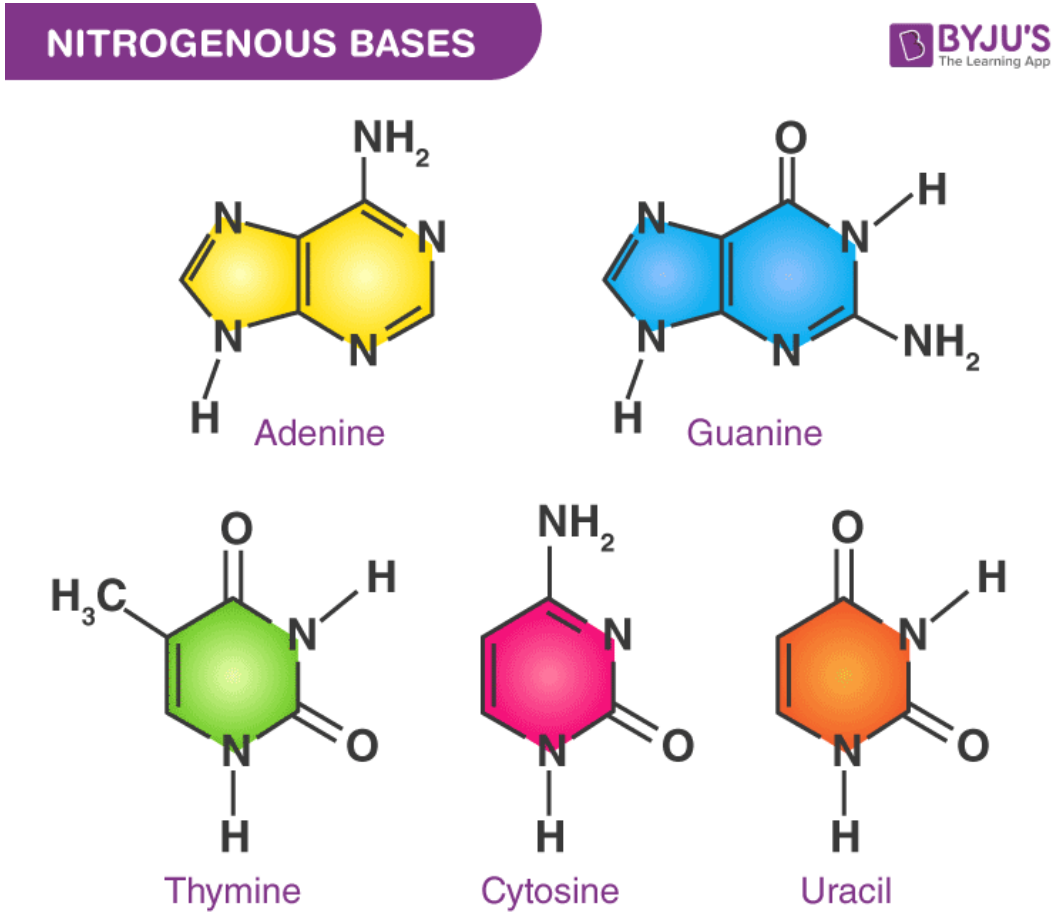
Understand the differences between bases, nucleosides, and nucleotides
The bases may be double ringed (purines) or single-ringed (pyrimidines)



https://microbenotes.com/nucleic-acids-nucleosides-and-nucleotides/



Purine = Pyrimidine + Imidazole Ring



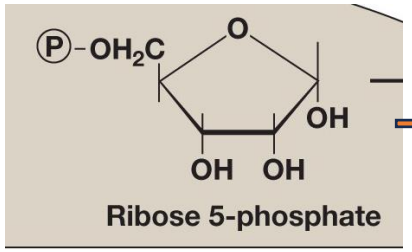
Q: Which bases/sugars are included in DNA/RNA?
A,G,T,C – DNA bases
A,G,C,U – RNA bases
Sugar: a pentose
Ribose – RNA
Deoxyribose - DNA

Purines and pyrimidines: diet vs synthesis

- We certainly get them from diet.
- Few of the purines and pyrimidines supplied by diet are used! We *degrade nearly all the dietary nucleotides* in the GI tract.
- However, we can synthesize them by *de novo synthesis*, or we may *salvage* them.
- Therefore, we will discuss their *synthesis, salvage, degradation, and clinical implications*.

Overview of purine de novo synthesis: two major blocks

Ribose 5-Phosphate (R5-P, or **RP**), provided by the Pentose 5-Phosphate pathway



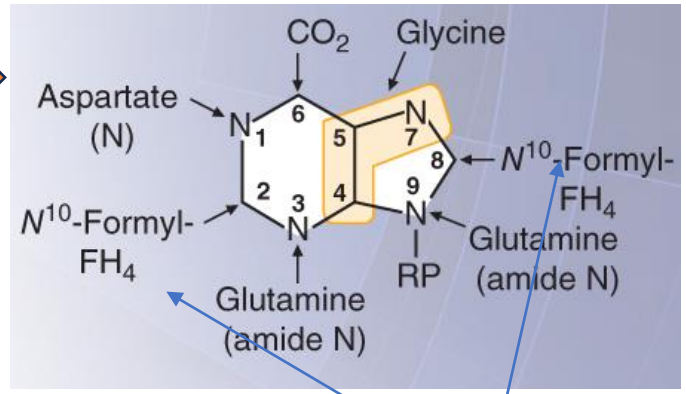
L, 22.6

Step-by-step addition of atoms/groups to build up the base on the sugar. See next slide for details!

Block 1: RP => IMP

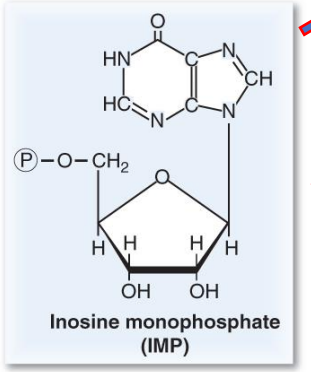
Atoms/groups donated by amino acids, CO_2, and FH_4

M, 39.1



Block 2: IMP => AMP/GMP

The base in IMP is hypoxanthine.



M, 39.6

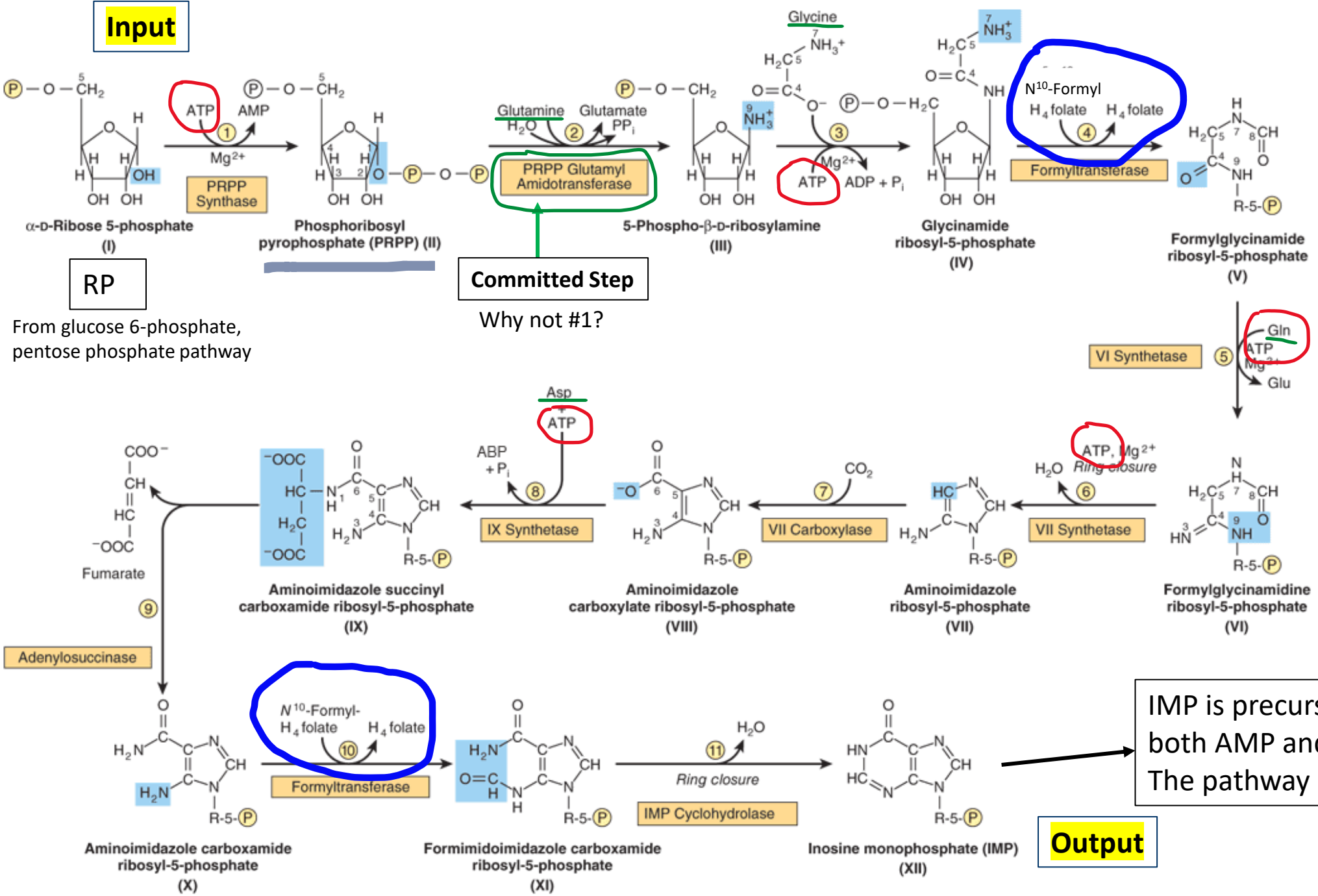
- AMP
  - ATP, RNA
  - dATP, DNA
- GMP
  - dGTP, DNA
  - ATP, RNA

Most de novo synthesis occur in the liver; the brain, neutrophiles, and cells of the immune system may also synthesize significant amounts of nucleotides

The liver may also break them down to bases and nucleosides for transport to other tissues (full nucleotides cannot be easily transported…)

Formyl Tetrahydrofolate (FH_4) is used as a primary one-carbon group carrier (INTG)

Amino acids are used as precursors! Yes, amino acids are used to make nucleic acids



RP

From glucose 6-phosphate, pentose phosphate pathway

Committed Step

Why not #1?

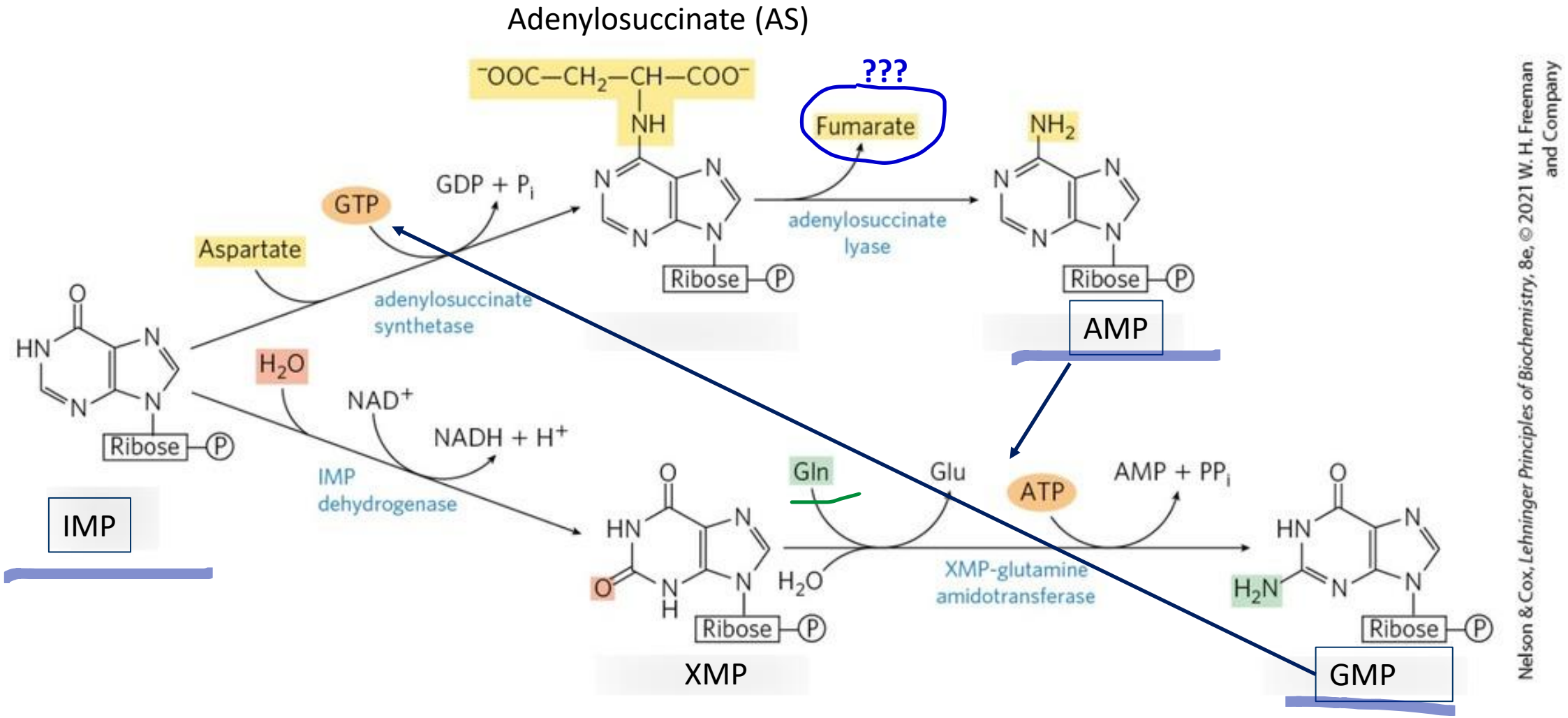
IMP synthesis:

- **many reactions**
- **Input: RP, Output: IMP**
  - **committed step (#2)**
- **energy (ATP) required**
- **donated atoms or entire groups are added to build the rings**
- **amino acids are needed**
- **tetrahydrofolate (N^{10}-formyl) is used** as a primary one-carbon group carrier!

IMP is precursor for both AMP and GMP! The pathway bifurcates.

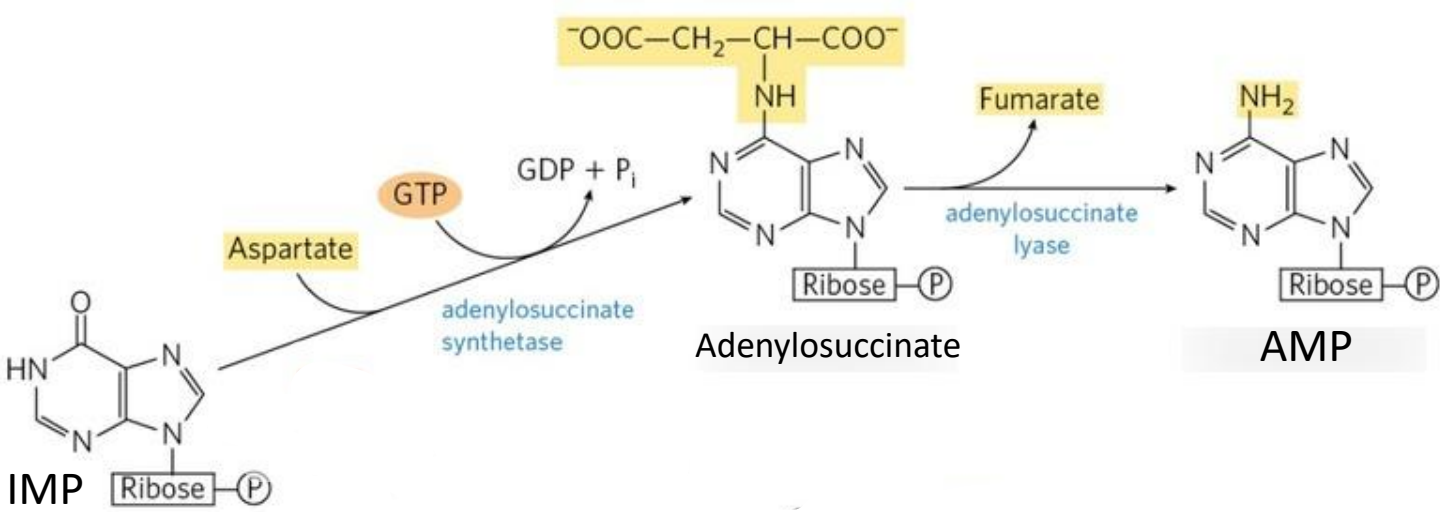
IMP to AMP/GMP bifurcation: two steps for each

Energy (ATP) and nitrogen (amino acids) are needed again

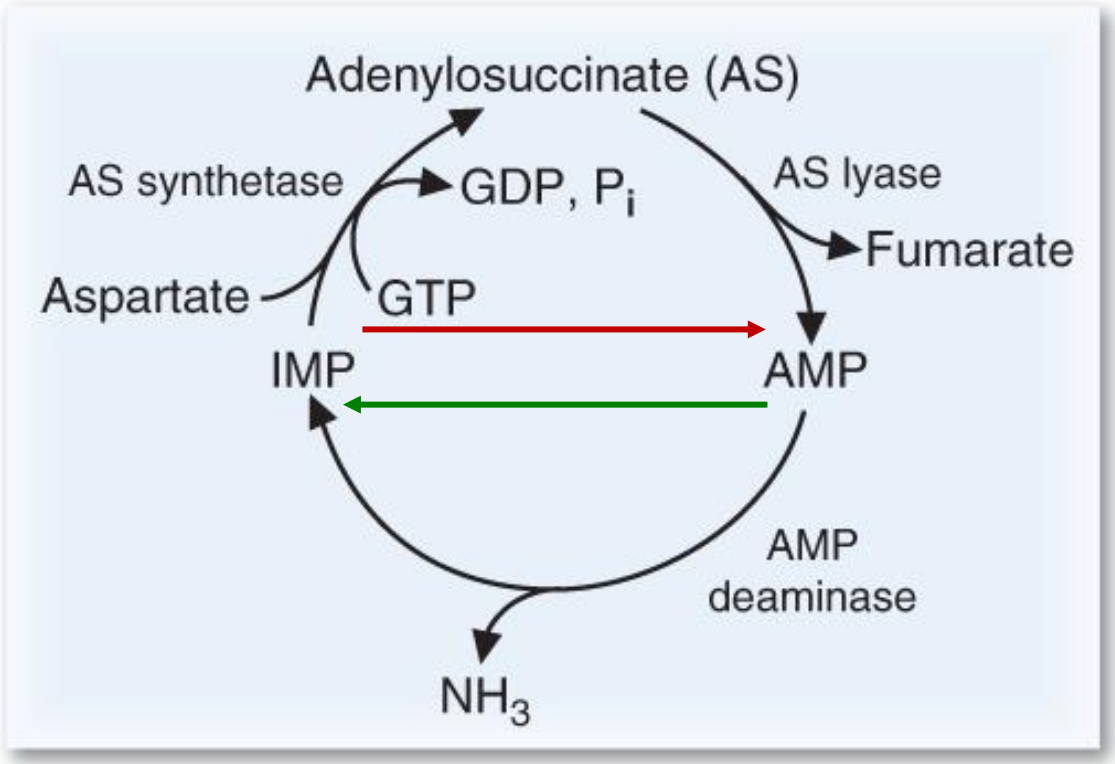


XMP: Xanthosine monophosphate: not found in DNA, precursor of other purines; found in anticodon of tRNA (enables wobble base pairs to form); **AMP and GMP are further used for ATP, GTP, dATP, dGTP, RNA, and DNA; AMP requires GTP, and GMP requires ATP; cross regulation is enabled**

What is the deal with the fumarate?



M, 39.12



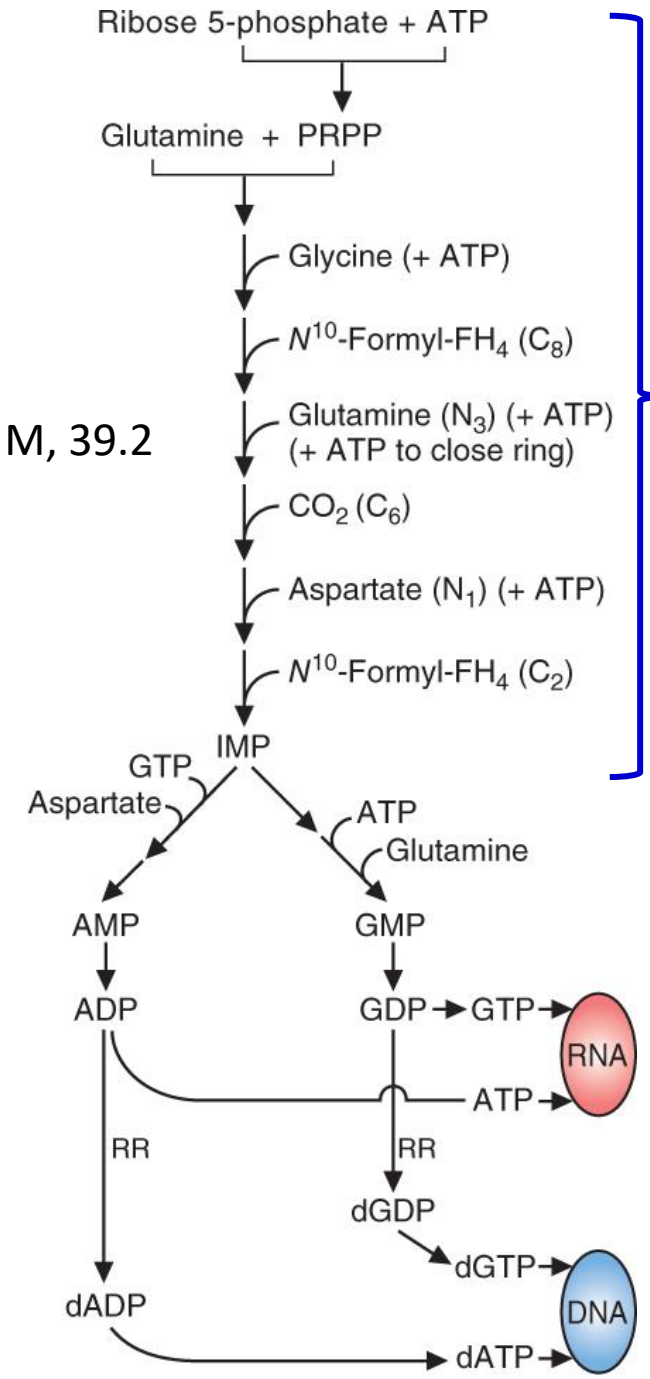
More on IMP-AMP pathway:
MSKS connection

The purine nucleotide cycle. Using a combination of biosynthetic and salvage enzymes, the net effect is the conversion of aspartate to fumarate, with plays an anaplerotic role in the muscle.

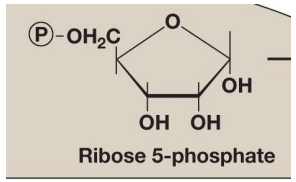
Anaplerosis: provides intermediates for other pathways

Under conditions in which the muscle must generate energy, the fumarate derived from the purine nucleotide cycle can be used anaplerotically to replenish TCA-cycle intermediates and to allow the cycle to operate at high speed. Deficiencies in enzymes of this cycle lead to muscle fatigue during exercise.

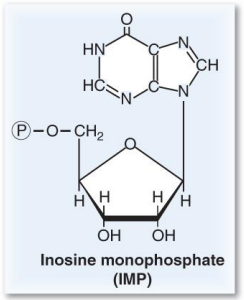
Purine-De Novo:Simplified Diagram



M, 39.2



Block I:
RP -> IMP



Block I: Start with RP, end with IMP (a ribose nucleotide)

Block II: Start with IMP (ribonucleotide); First bifurcation:

Yield AMP and GMP (ribonucleotides)

AMP and GMP are converted to ADP/GDP by monophosphate kinases

Next, new bifurcation occurs for each pathway:

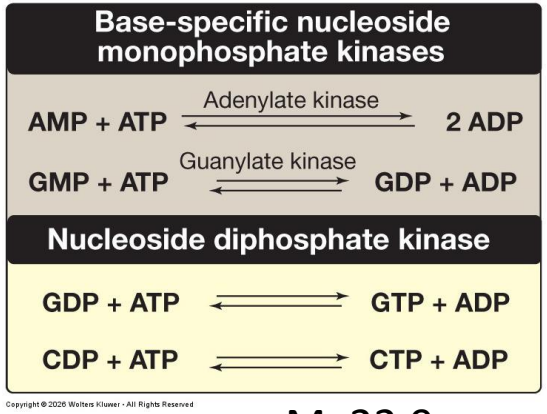
ADP/GDP are converted ATP and GTP by diphosphate kinases to be used as precursors for RNA

or

ADP/GDP are converted to dADP/dGDP (deoxyribonucleotides) by Ribonucleotide Reductase (RR)

and

dADP/dGDP are converted to dATP and dGTP by diphosphate kinases to further be used as precursors for DNA



M, 22.9

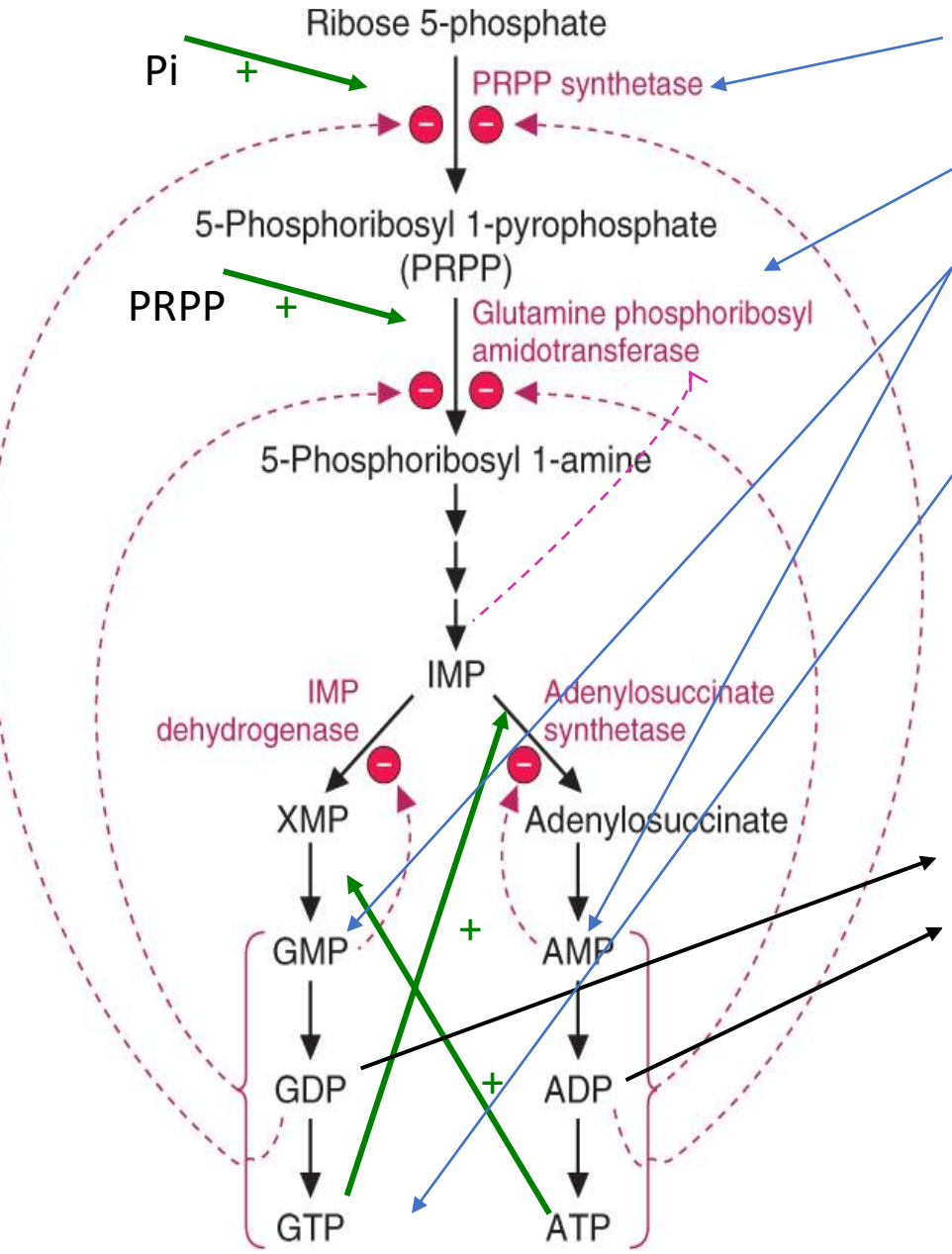
Block II:
IMP -> Nucleotides -> Nucleic Acids

Conversion of nucleoside monophosphates to di-phosphates is catalyzed by base-specific (not sugar-specific) monophosphate kinases

Conversion of di- to tri-nucleoside phosphates is carried out by non-base-specific diphosphate kinases (also used for pyrimidines!)

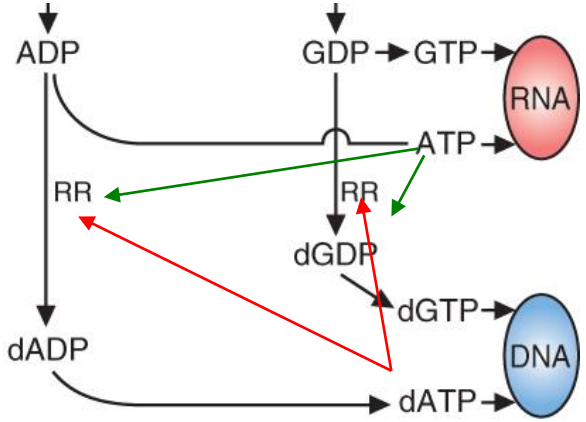
RR acts on both purines and pyrimidines (recognizes diphosphates, and acts on sugars – not bases).

Purine synthesis regulation: + = activators, - = inhibitors

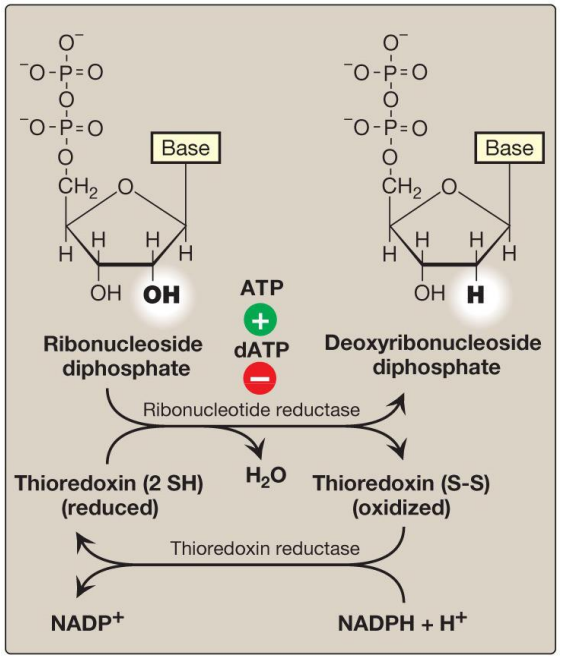


L, 39.9

- PRPP synthetase has two distinct, inhibitory allosteric sites, one for ADP, the other for GDP.
- Glutamine (glutamyl) phosphoribosyl amidotransferase has AMP and GMP binding sites; the monophosphates are the most important, although di- and tri-phosphates, and IMP will also inhibit the enzyme. This is the committed step (generally, the site of major regulation)!
- GMP and AMP act as negative effectors at branch points (feedback inhibition). Adenylosuccinate synthetase is inhibited by AMP; IMP dehydrogenase is inhibited by GMP.
- The synthesis of AMP is dependent on GTP (of which GMP is a precursor); the synthesis of GMP is dependent on ATP (which is made from AMP). High ATP → GMP will be made; high GTP → AMP synthesis will take place. This balances the products.



For DNA synthesis,
RR is regulated by ATP and dATP
RR is also used for pyrimidines!



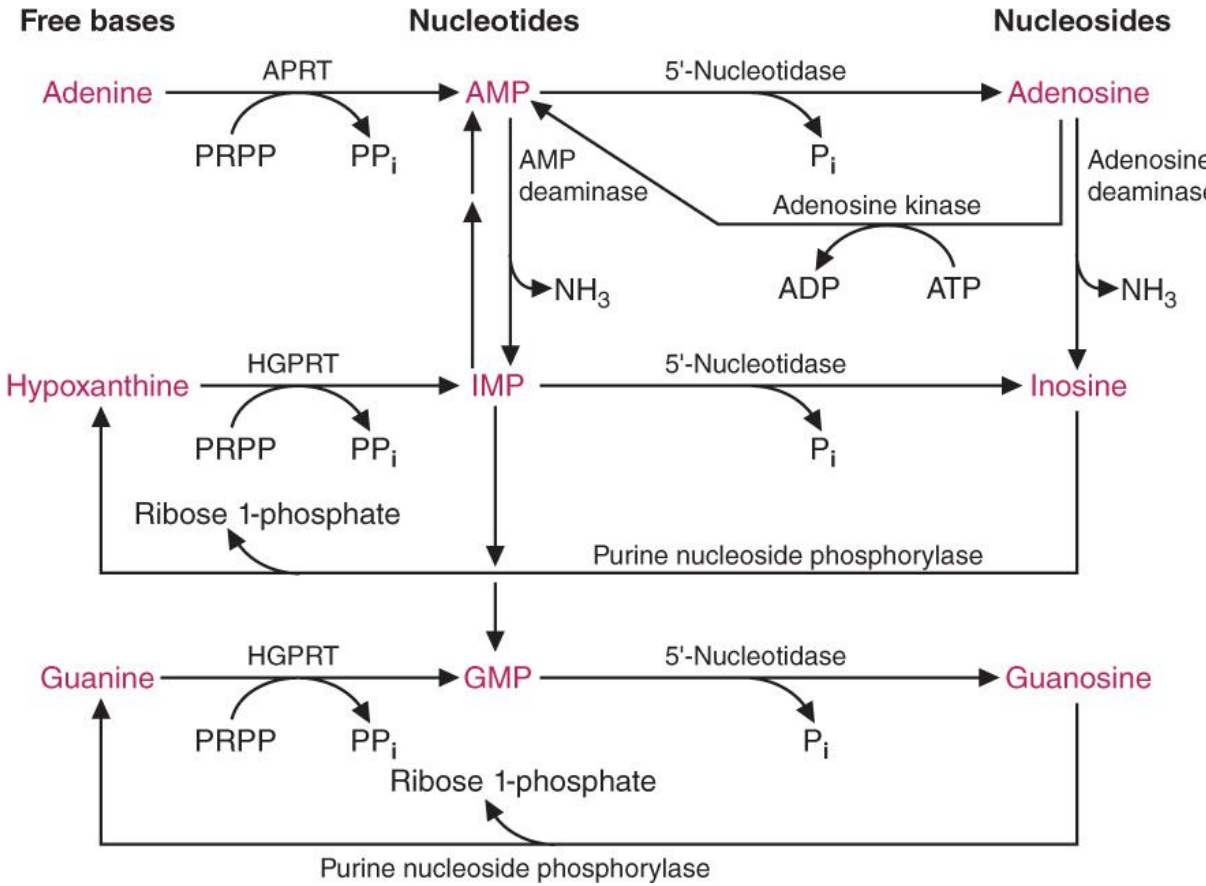
M, 22.12

-Substrate-induced activation
-Feedback inhibition (major)

Purine Salvage Pathway

- **Key: the salvage pathway allow free bases, nucleosides, and nucleotides to be interconverted.**
- Salvage is the major form of nucleotide generation in certain cells, such as lymphocytes.

- Nucleotides can be converted to nucleosides by 5`-nucleotidase.
- Free bases can be made from nucleosides by **purine nucleoside phosphorylase (PNP).**
- The free bases may be salvaged or degraded, depending on cellular needs.
- Deamination of adenines occurs with AMP-deaminase and **adenosine deaminase (ADA).** Of the purines, only adenosine can be directly phosphorylated back to a nucleotide, by adenosine kinase.
- The purine bases hypoxanthine and guanine react with PRPP to form the nucleotides inosine and guanosine monophosphate, respectively. The enzyme that catalyzes the reaction is **hypoxanthine-guanine phosphoribosyltransferase (HGPRT).**
- Adenine directly forms AMP in a reaction with PRPP catalyzed by adenine phosphoribosyltransferase (APRT).



L, 39.10

Salvage is also regulated by feedback inhibition presented by IMP, AMP, GMP!
It is well integrated with the ***de novo*** **synthesis (i.e., PRPP, IMP, GMP, AMP)** and degradation (see later)!

Degradation of dietary nucleotides

-Few of purines and pyrimidines supplied by diet (from RNA/DNA) are utilized. The nucleic acids are degraded in the GI tract

-Denaturation by the low pH in the stomach prepare them for the attack of pancreatic nucleases that hydrolyze them to oligonucleotides

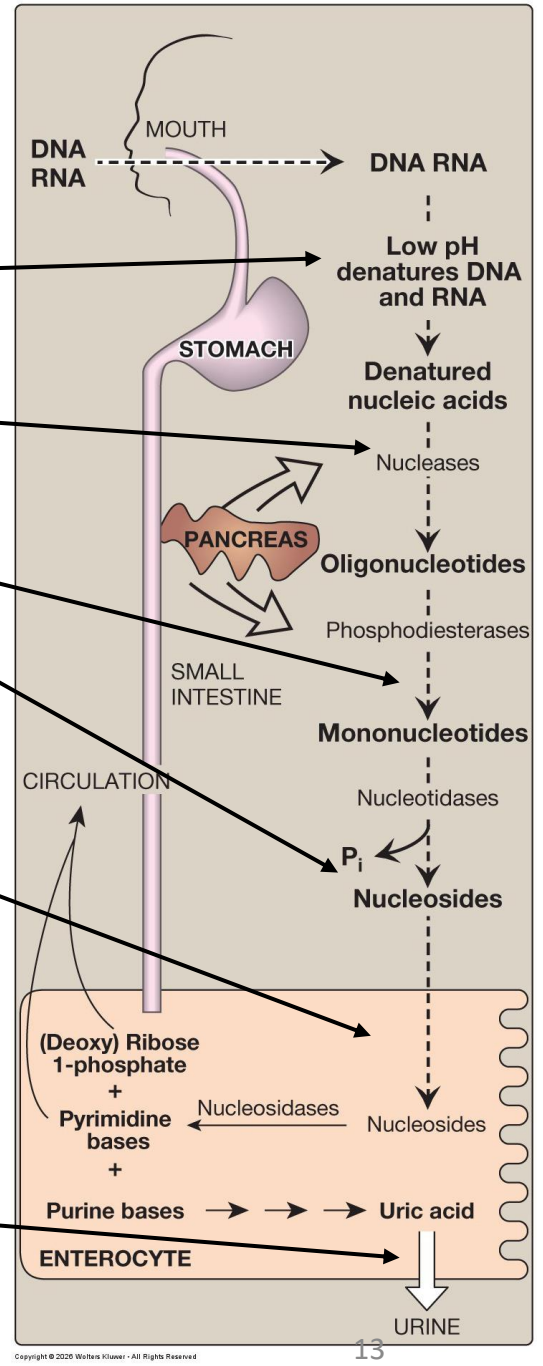
-Pancreatic phosphodiesterases produce a mixture of 3′- and 5′-mononucleotides

-At the intestinal mucosal surface, nucleotidases remove the phosphate groups and release nucleosides

-Nucleosides are taken into enterocytes by sodium-dependent transporters and degraded by nucleosidases (nucleoside phosphorylases) to free bases plus (deoxy)ribose 1-phosphate

-Dietary purine bases are not used to any appreciable extent for the synthesis of tissue nucleic acids. Instead, they are degraded to uric acid in the enterocytes. Uric acid is the end product of intestinal purine degradation. Dietary uptake of pyrimidine bases is also low

Most of the uric acid enters the blood and is eventually excreted in the urine



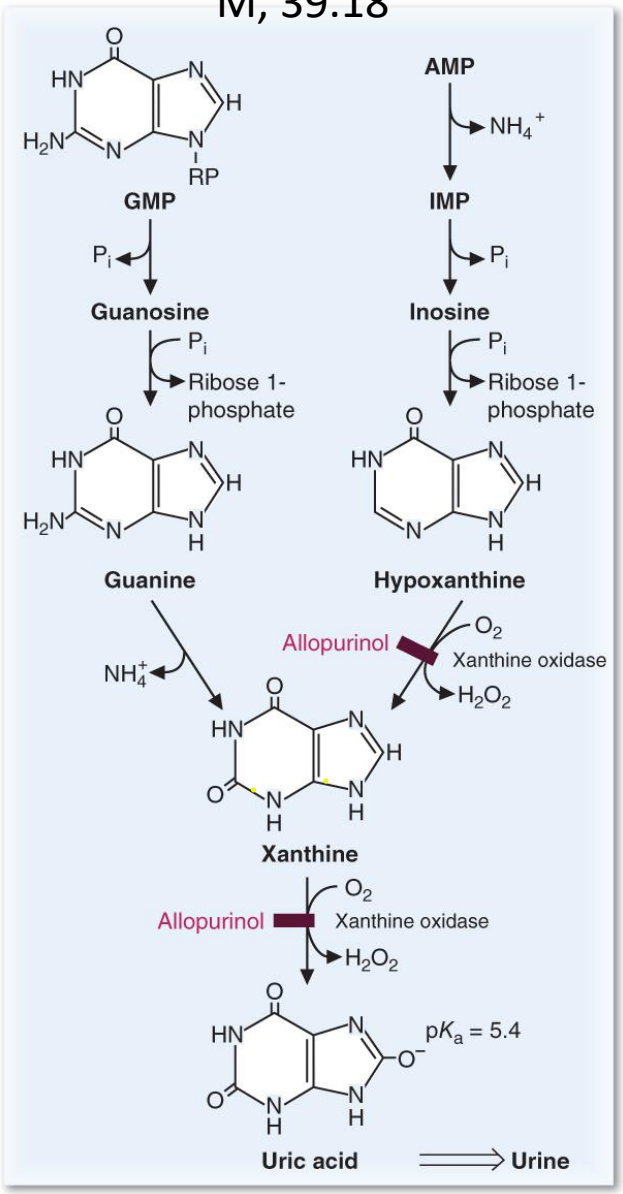
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L, 22.14

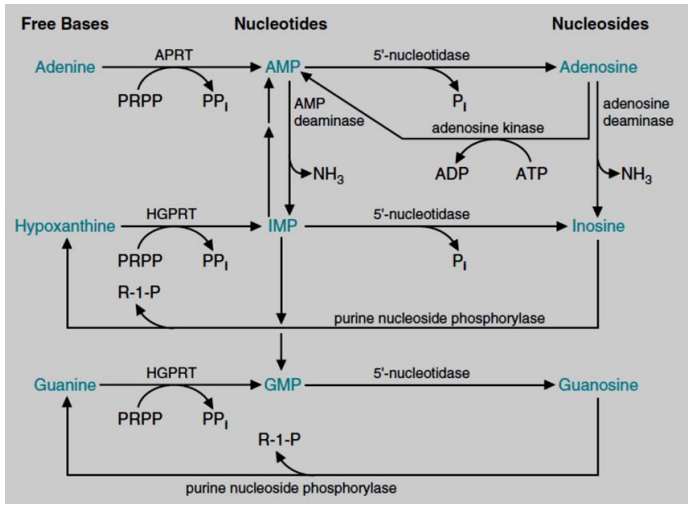
Purine Degradation Pathways:

Integrated with synthesis/salvage

M, 39.18



M, 39.10



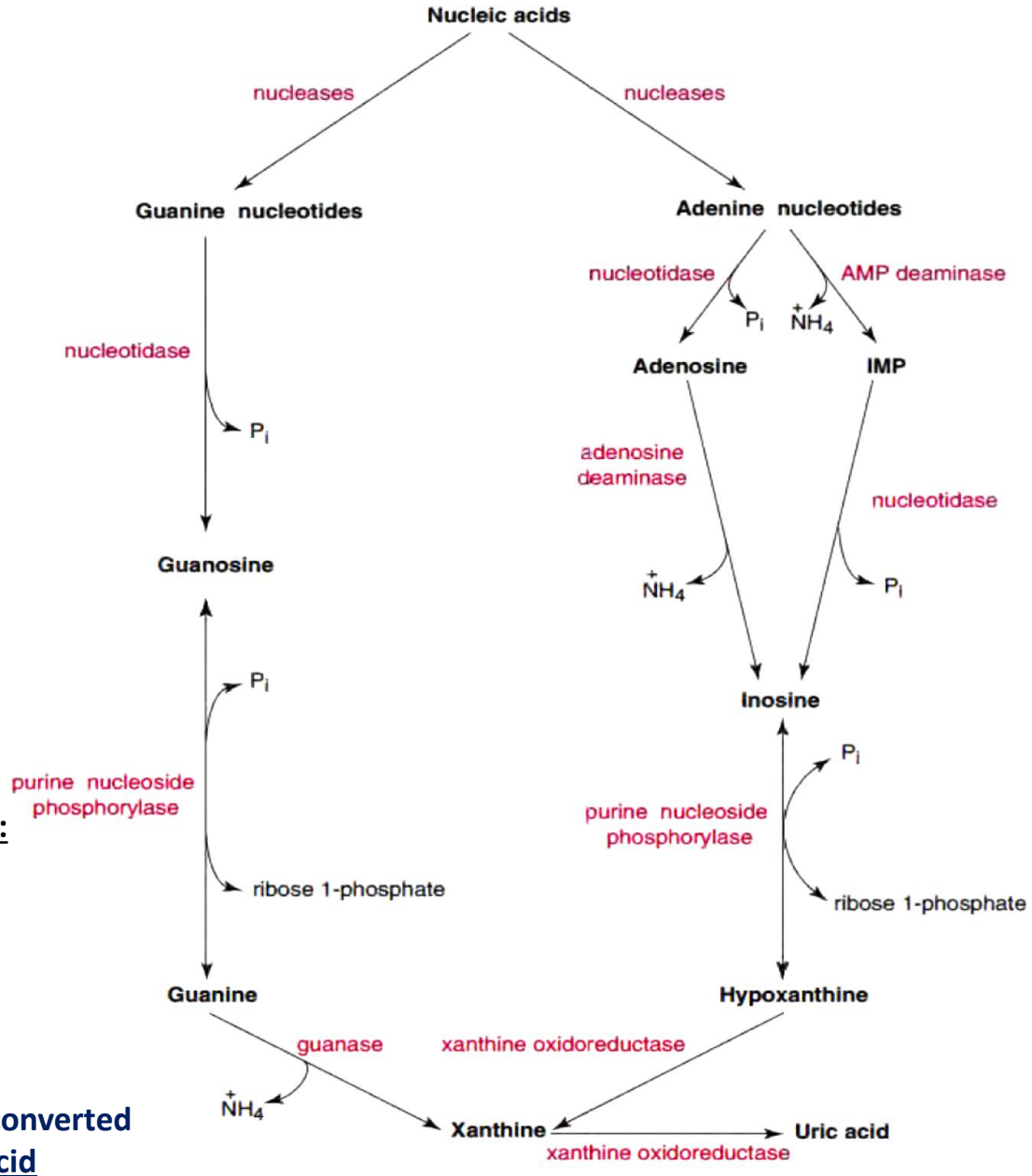
Salvage

<u>Enzymes from salvage are also used for degradation:</u>
AMP deaminase, ADA, nucleotidase, PNP

<u>Hypoxanthine and Guanine are common to salvage/degradation pathways</u>
<u>PRPP is common to synthesis</u>

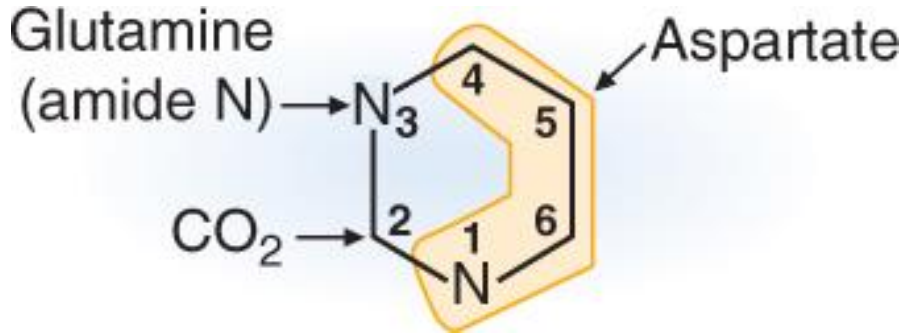
All the nucleotides undergoing degradation will be converted to Xanthine. XO/XOR will convert Xanthine to <u>uric acid</u>

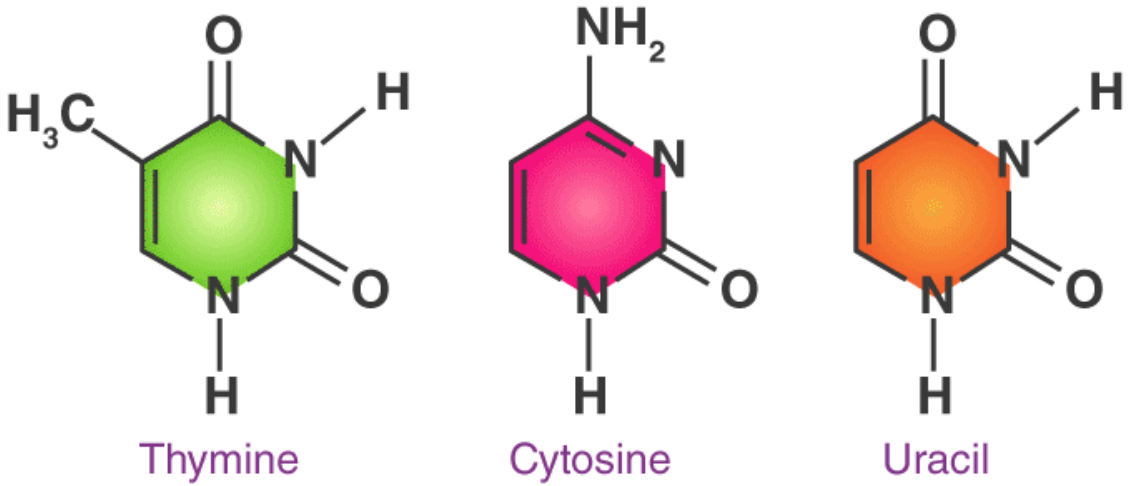
Dietary purine nucleotide are degraded in enterocytes
Other (excess) are degraded in the liver



Degradation

Pyrimidines: de novo synthesis



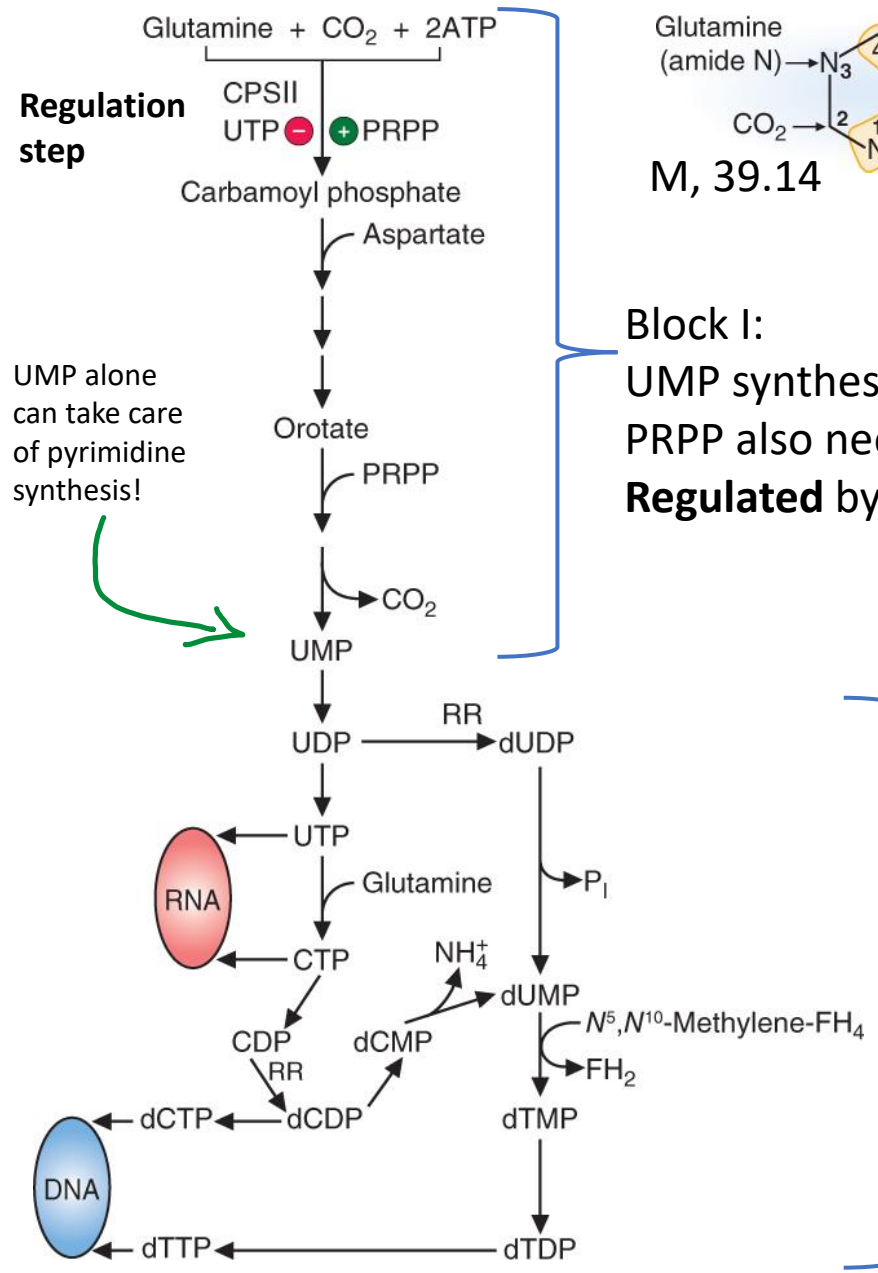


M, 39.14

Unlike the synthesis of the purine ring, which is constructed on a pre-existing ribose 5-phosphate, **the pyrimidine ring is fully synthesized** before being attached to **ribose 5-phosphate, which is donated** by **PRPP**. The sources of the atoms in the pyrimidine ring are glutamine, CO_2, and aspartate

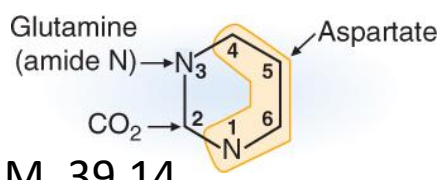
Pyrimidine Synthesis at a Glance

You should know carbamoyl phosphate from urea cycle (uses CPS I)! Pyrimidine synthesis uses CPS II. Here are the differences.



RR: ribonucleotide reductase

M, 39.13



M, 39.14

Block I:
UMP synthesis
PRPP also needed!
Regulated by **UTP/PRPP**

Variable	CPS I	CPS II
Cellular location	Mitochondria	Cytosol
Pathway involved	Urea cycle	Pyrimidine synthesis
Source of nitrogen	Ammonia	γ-Amide group of glutamine
Regulators	Activator: N-acetyl-glutamate	Activator: PRPP Inhibitor: UTP

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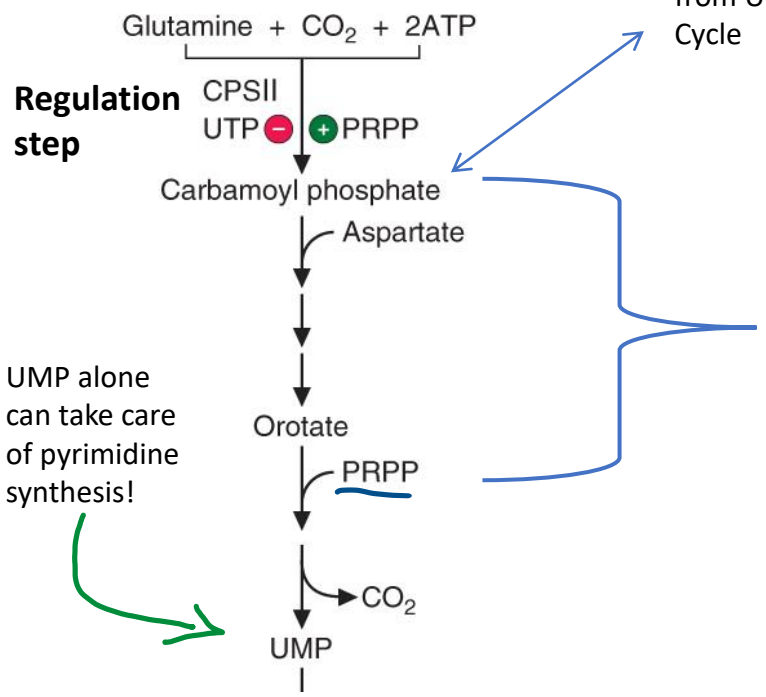
L, 22.20

Summary of the differences between **carbamoyl phosphate synthetase (CPS) I and II**.
NAG is an ON-OFF switch for CPS I.

Block II:
Conversion to RNA/DNA precursors
Only UMP is needed to start it
Therapeutic implications when Block I is non functional

RR plays the same role as in purines (ribonucleotide conversion to deoxyribonucleotides)

M, 39.13



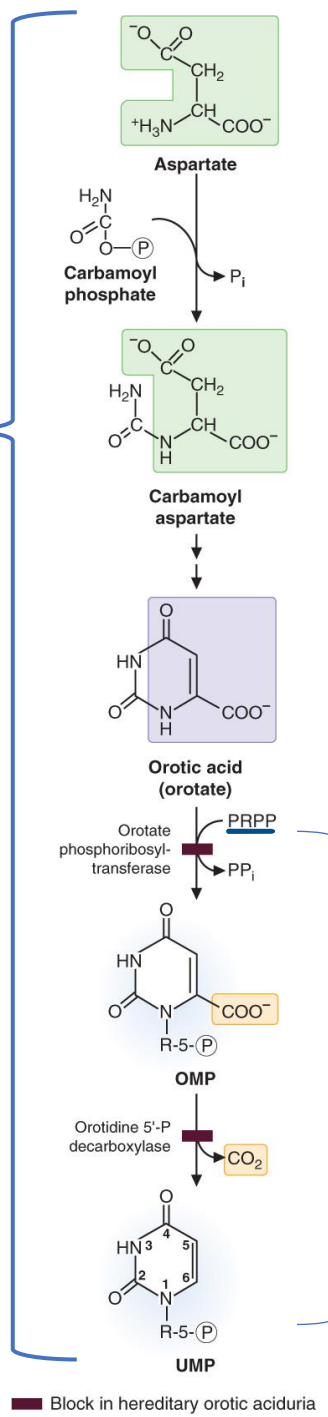
Regulation step

May come from Urea Cycle

UMP alone can take care of pyrimidine synthesis!

Pyrimidine Synthesis, Block I - Explained

Defects in ornithine transcarbamylase (OTC) of the urea cycle promote pyrimidine synthesis because of the increased availability of carbamoyl phosphate. Purine synthesis is not affected.



In mammals, CPSII, Aspartate transcarbamoylase, and Dihydroorotase are located on the same same polypeptide – CAD

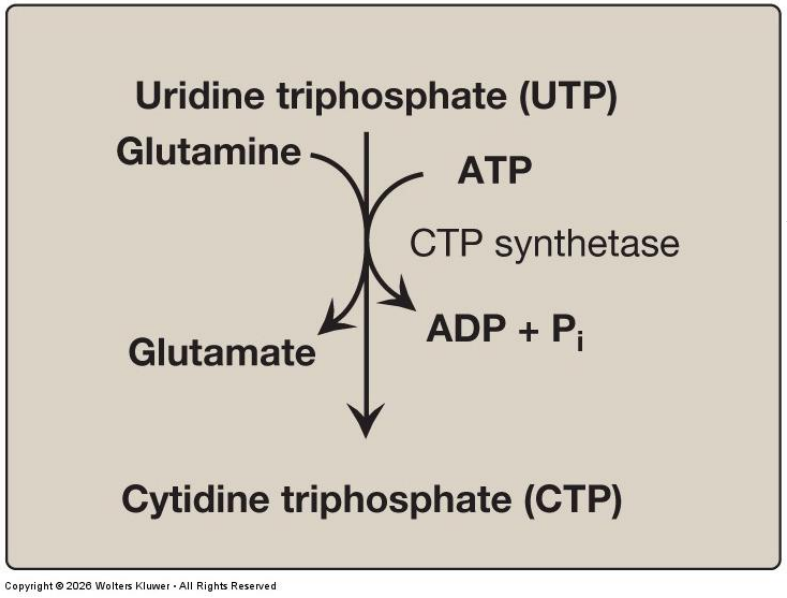
PRPP is used as Ribose-Phosphate donor

Orotate phosphoribosyl transferase and orotidylic acid decarboxylase are both located on another polypeptide, UMP synthase

M, 39.15

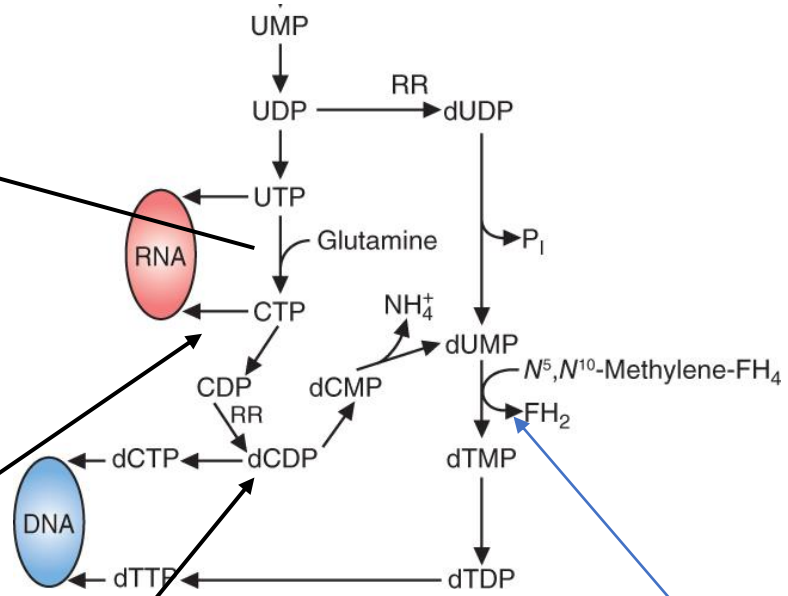
Block II: RNA/DNA precursors

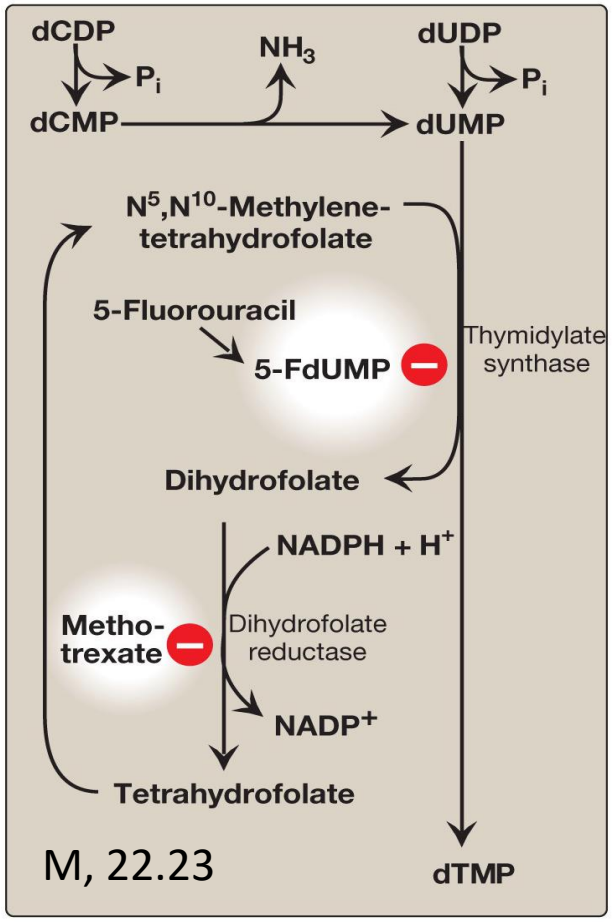
RNA: UTP, CTP
DNA: dCTP and dTTP, RR needed



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M, 22.22





M, 22.23

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Left:UTP is used for RNA or to produce Cytidine Triphosphate (CTP) by amination (CTP synthetase), with glutamine providing the nitrogen.

CTP is used for RNA but some is dephosphorylated to CDP, which is a substrate for ribonucleotide reductase (RR).

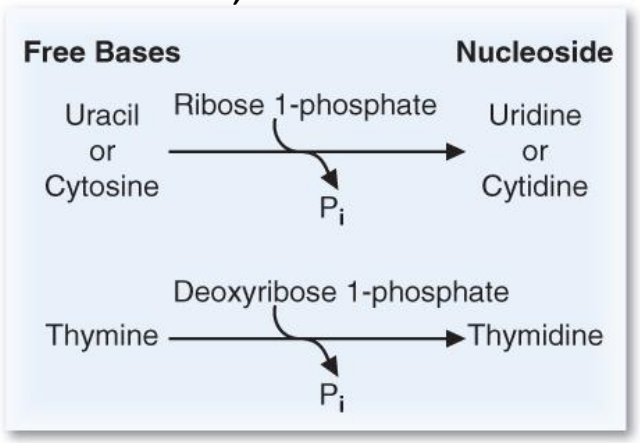
dCDP can be phosphorylated to dCTP for DNA synthesis or dephosphorylated to dCMP that is further deaminated to dUMP and converted to dTMP/dTDP/dTTP (for DNA).

Right: dUMP (from either dUDP or dCDP) is converted to dTMP by thymidylate synthase, which uses N^5,N^{10}-methylene-THF as the source of the methyl group.

We will present this pathway later in the context of drugs!!!

Pyrimidine salvage: free bases – nucleosides - nucleotides

M, 39.16



Step 1

M, Table 39.2

Salvage Reactions for Conversion of Pyrimidine Nucleosides to Nucleotides

ENZYME	REACTION
Uridine-cytidine kinase	Uridine + ATP → UMP + ADP
	Cytidine + ATP → CMP + ADP
Deoxythymidine kinase	Deoxythymidine + ATP → dTMP + ADP
Deoxycytidine kinase	Deoxycytidine + ATP → dCMP + ADP

Step 2

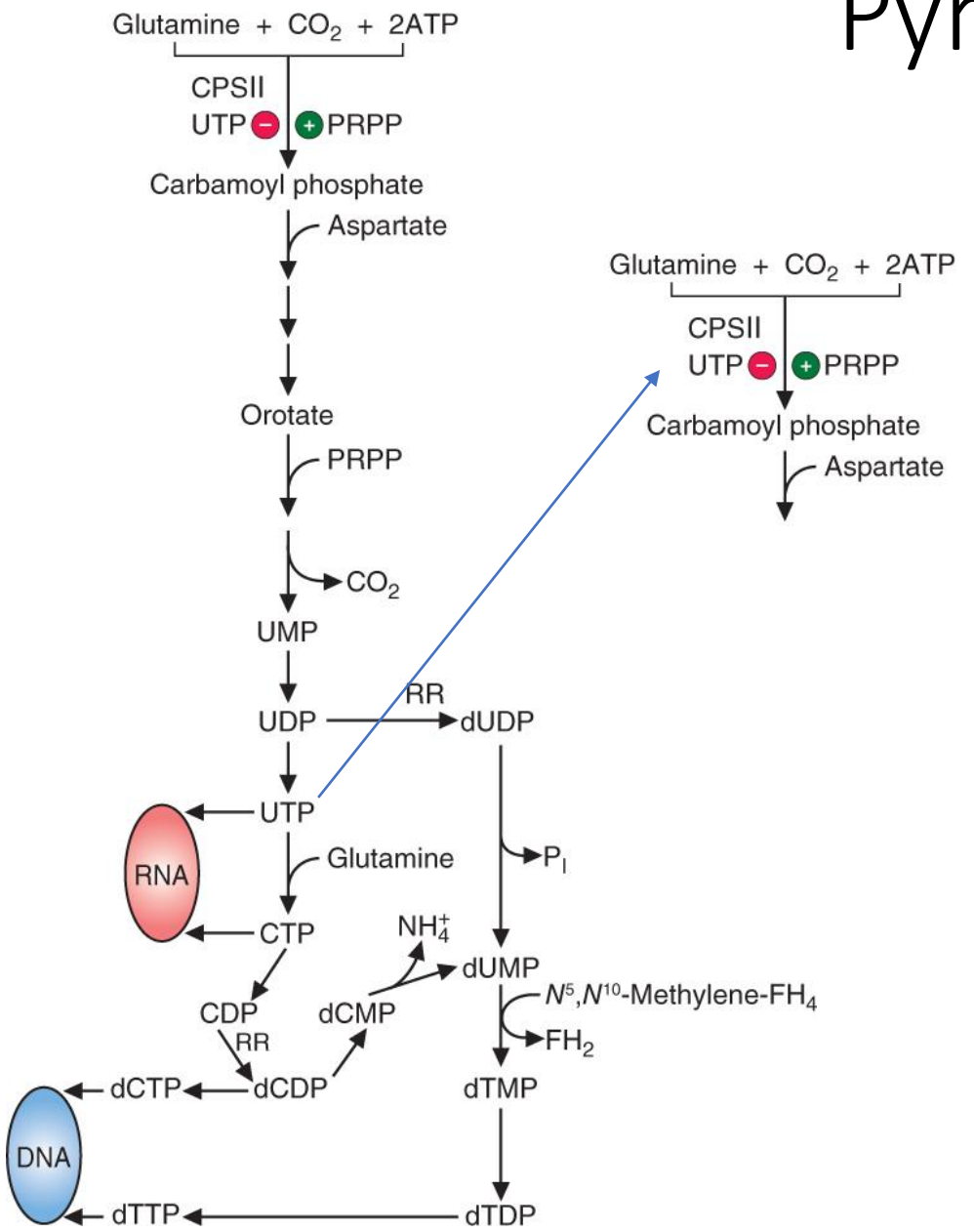
Pyrimidine bases are normally salvaged by a two-step route:

Step 1: a relatively nonspecific pyrimidine nucleoside phosphorylase converts the pyrimidine bases to their respective nucleosides. The first phosphorylase (adds the ribose) has high affinity for uracil, less for cytosine, and almost nothing for thymine. The second phosphorylase adds a deoxyribose to thymine.

Step 2: more specific nucleoside kinases react with the nucleosides, forming nucleotides.

IMPORTANT: Uridine can be salvaged; important for clinical implications. We may supply it if needed, and we may synthesize all the pyrimidines from uridine.

Pyrimidine synthesis regulation



M, 39.13

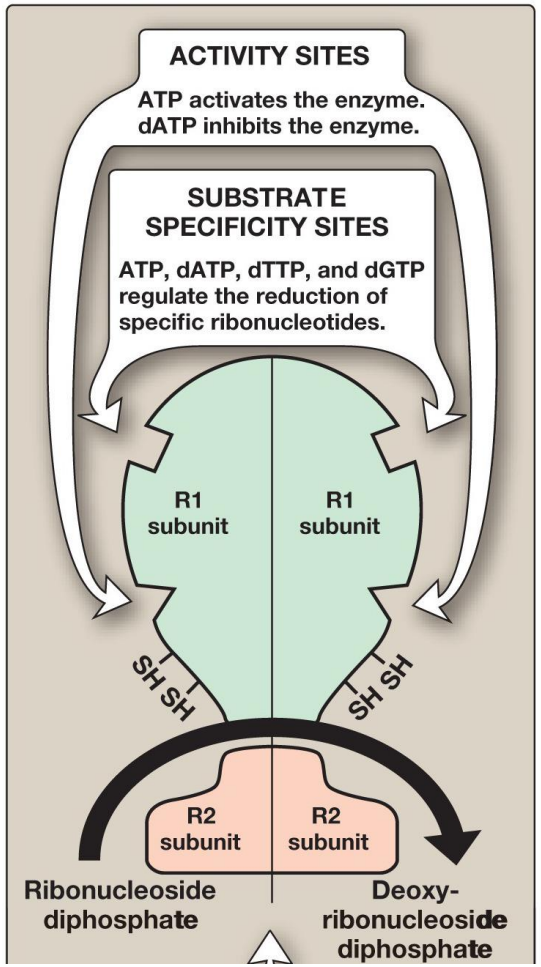
Two major regulation sites, pathway (specific) and Ribonucleotide Reductase (common for purines and pyrimidines)

Pathway regulation: The regulated step of pyrimidine *de novo* synthesis in humans is CPSII. The enzyme is inhibited by UTP and activated by PRPP.

Thus, as pyrimidines decrease in concentration (as indicated by UTP levels), CPSII is activated, and pyrimidines are synthesized.

RR regulation is more intricate (see next)

Ribonucleotide reductase (RR) regulation: purines and pyrimidines



L, 22.13

Goal: maintain a balanced supply of deoxynucleotides for DNA synthesis

Two subunits (R1, alpha; R2, beta); each set of subunits forms homodimers.

Contains sets of allosteric sites, one set controlling the activity of the enzyme and the other set controlling the substrate specificity of the enzyme.

Activity sites: **ATP (purine)** bound to the activity site **activates the enzyme**; deoxyadenosine triphosphate **(dATP, purine)** bound to this site **inhibits** the enzyme. **Increased dATP is toxic** because prevents DNA synthesis (see later ADA deficiency)

Substrate specificity is more intricate (and connected to pyrimidines). **ATP (purine) bound to the substrate specificity** site **activates** the reduction of **pyrimidines (CDP and UDP),** to form deoxycytidine diphosphate (dCDP) and deoxyuridine diphosphate (dUDP).

dUDP is not used for DNA synthesis; rather, it is used to **produce dTMP**. Once dTMP is produced, it is phosphorylated to dTTP, which then binds to the substrate specificity site and induces the reduction of GDP.

As **dGTP accumulates**, it replaces **dTTP** in the substrate specificity site and allows ADP to be reduced to **dADP**. This leads to the **accumulation of dATP**, which **inhibits the overall activity** of the enzyme.

Note: RR is inhibited by hydroxyurea (which we will discuss in the Drugs section).

Degradation of Pyrimidines

- **The pyrimidine nucleotides are dephosphorylated, and the nucleosides are cleaved to produce ribose 1-phosphate and the free pyrimidine bases cytosine, uracil, and thymine.**

- **Note 1: the products of pyrimidine degradation are excreted in the urine or converted to CO_2, H_2O, and NH_4^+ (which forms urea). They do not cause major problems for the body because they are highly soluble!**

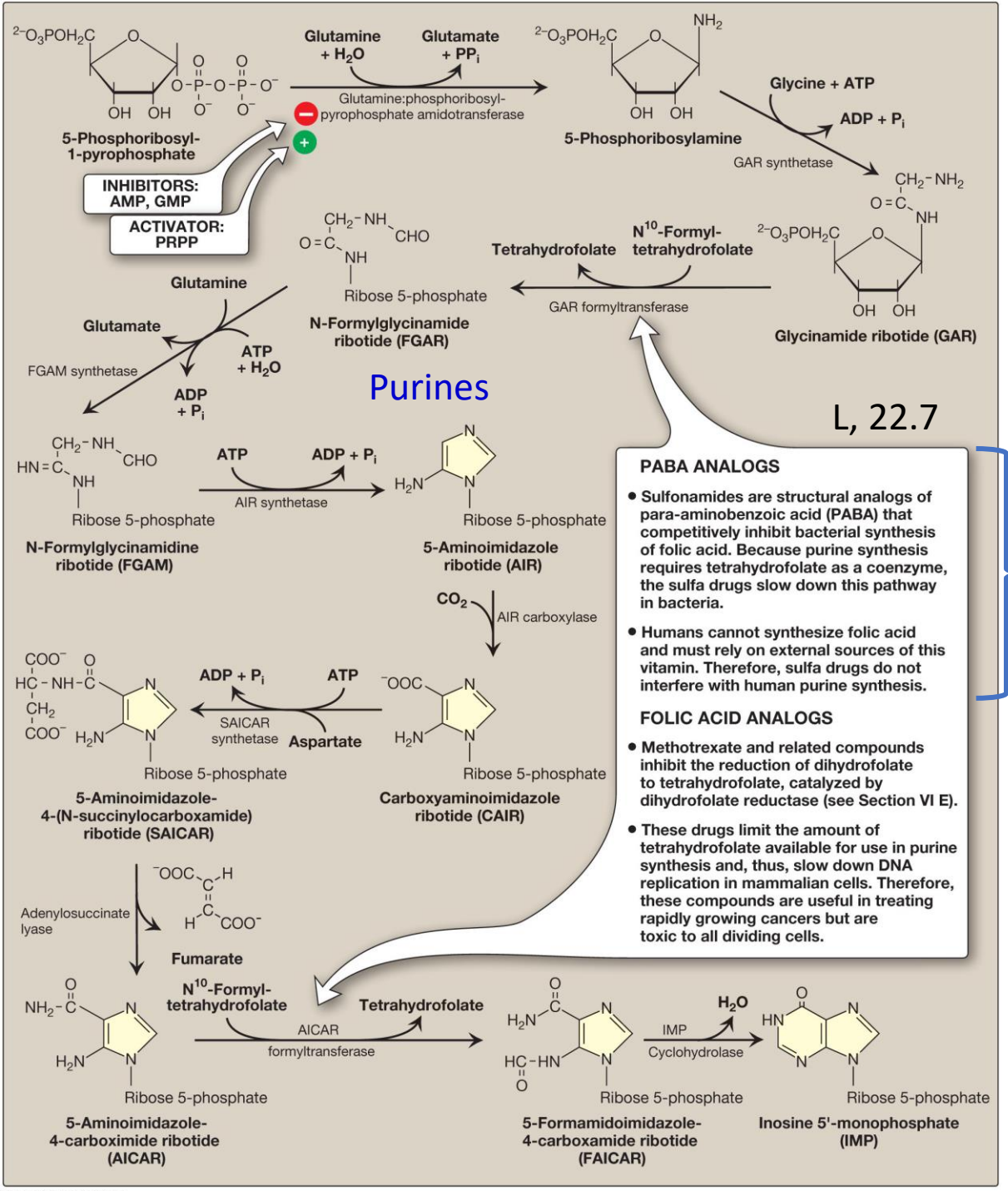
- **Note 2: Pyrimidine bases can be salvaged to nucleosides, which are phosphorylated to nucleotides.**

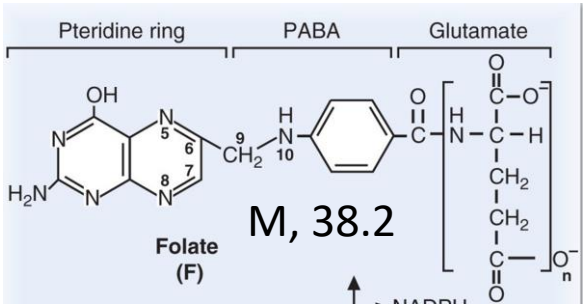
- **Note 3: The salvage of pyrimidine nucleosides is the basis for using uridine in the treatment of hereditary orotic aciduria (see later).**

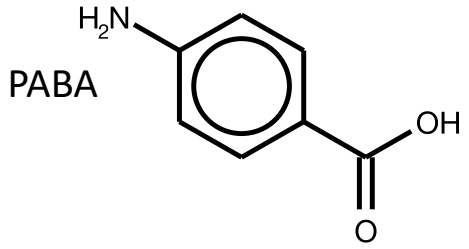
Clinical implications: Outline

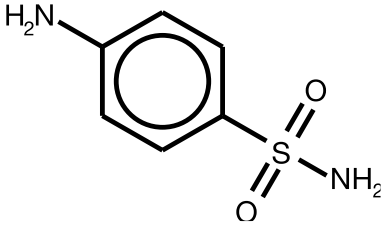
- **Purines/pyrimidines de novo synthesis: biochemical rationale of drugs**
- **Pyrimidines: orotic aciduria**
- **Purine salvage/degradation: Lesch-Nyhan syndrome, ADA/PNP deficiency**
- **Purine degradation: Hyperuricemia, Gout**

Antifolate Drugs: Structural PABA analogs







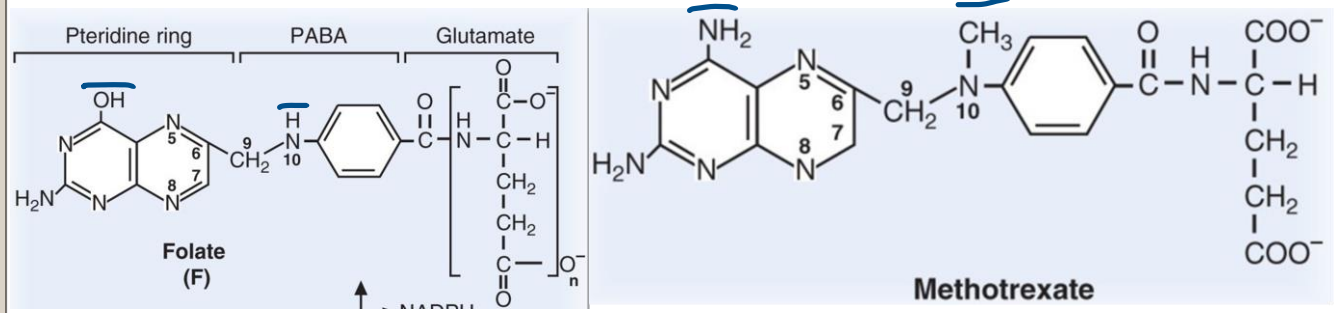


Folate => Dihydrofolate => Tetrahydrofolate

Tetrahydrofolate is key for de novo synthesis of both purines and pyrimidines; derived from **folic acid**; bacteria synthesize it by using **Para-aminobenzoic acid (PABA). PABA analogs (i.e., sulfonamides) block folate synthesis in bacteria and reduce proliferation. Antibiotic properties.**

Humans cannot synthesize folic acid (vitamin B9). Sulfa drugs do not interfere with nucleotide metabolism in humans.

Antifolate drugs: methotrexate

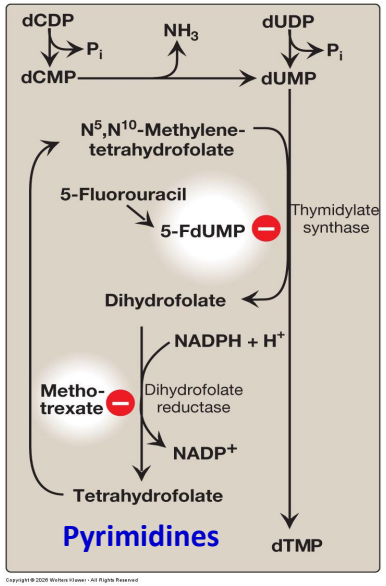


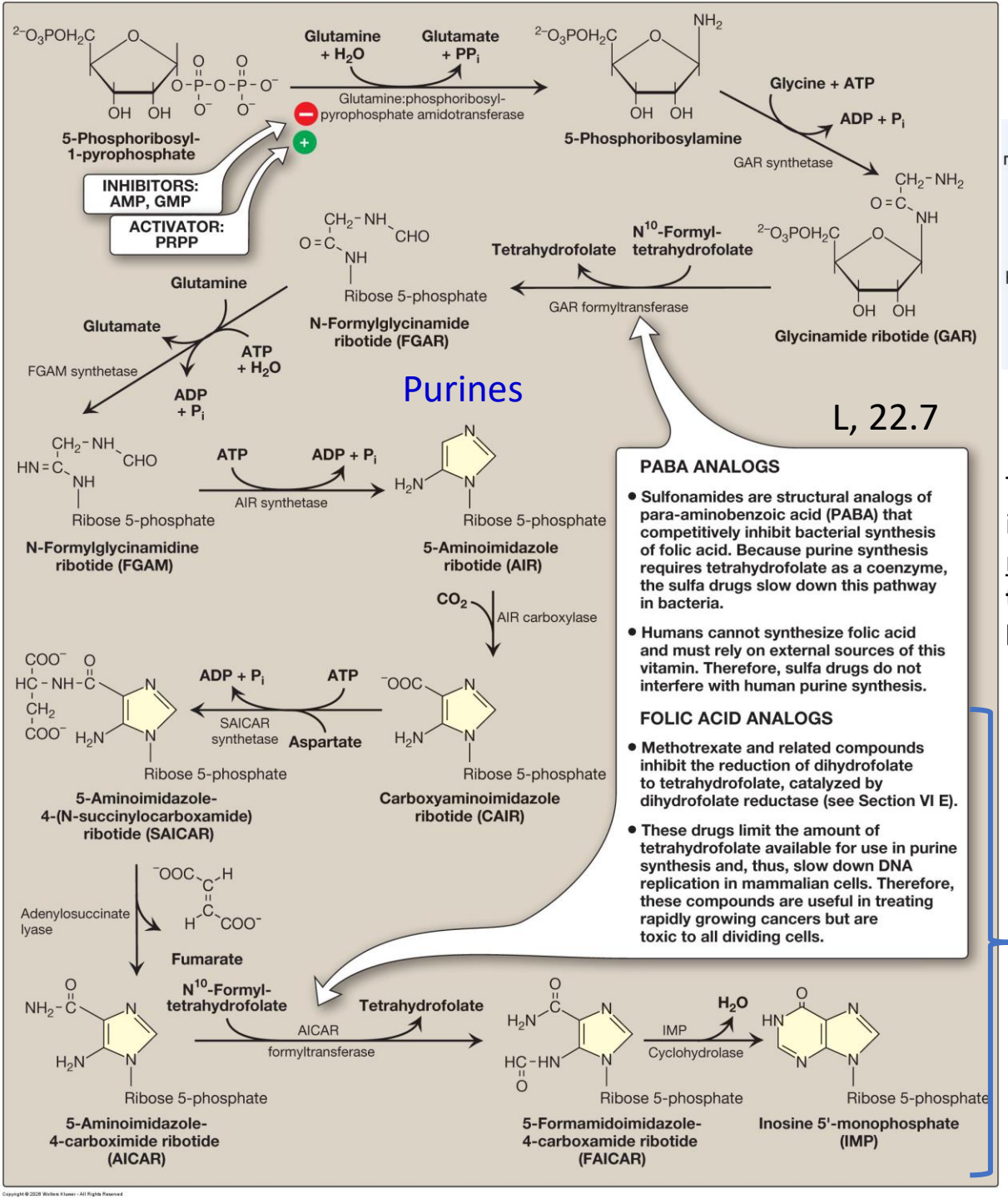
M, 38.2 M, page 855, Q

Methotrexate and related compounds are structural analogs of folic acid. They inhibit reduction of dihydrofolate to tetrahydrofolate – nucleotide synthesis is slowed down (affects purine and pyrimidines) – DNA replication is impeded – rapidly growing cells cannot proliferate. **Useful for treating cancer but they are toxic to all rapidly dividing cells (bone marrow, skin, GI tract, immune system, hair follicles).**

Methotrexate: inhibits the enzyme Dihydrofolate reductase (DHR, or DHFR); reduction of dihydrofolate to tetrahydrofolate is inhibited; purines (left) and pyrimidines (right) synthesis is affected; DNA synthesis slow down; **anticancer** drug.

Trimethoprim also inhibits DHR but mostly in bacteria – not in humans; used as an **antibiotic** (especially UTI), often in combination with sulfonamides.



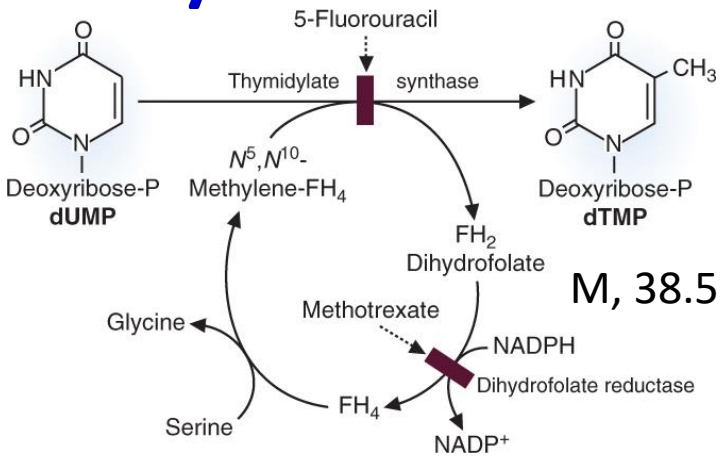


Clinical implications of dTMP synthesis: Drugs

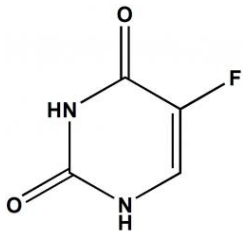
- The analog 5-fluorouracil (5-FU) serves as an antitumor agent.
  - 5-Fluorouracil is metabolically converted to 5-fluorodeoxyuridine monophosphate (5-FdUMP), which permanently inactivates the thymidylate synthase, making the drug a suicide inhibitor.
  - Due to decreased precursor availability, DNA synthesis and cell growth are inhibited

•A second metabolite (FUTP) integrates into developing RNA strands in place of normal uridine triphosphate (UTP). This creates defective, "fraudulent" RNA, which severely interferes with vital cellular processing and protein synthesis.
•A third metabolite (FdUTP) can be accidentally packed into DNA strands. This structural error causes severe DNA strand breaks and triggers cellular apoptosis (programmed cell death)

- The drug is used to treat cancer.



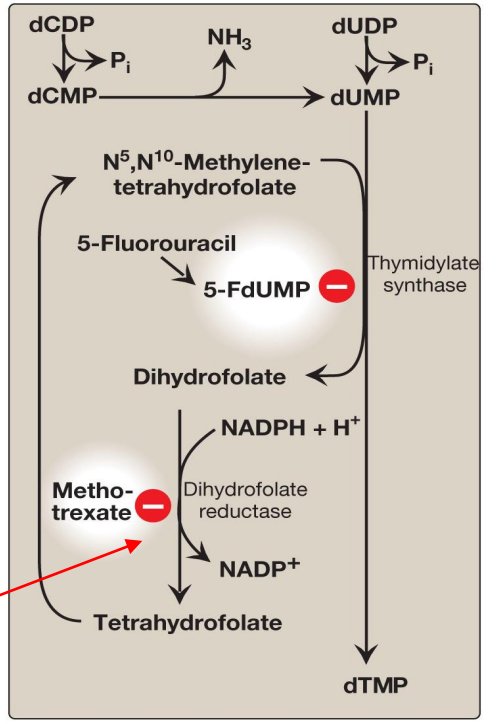
M, 38.5



5-Flourouracil

-inhibitor, bacteria

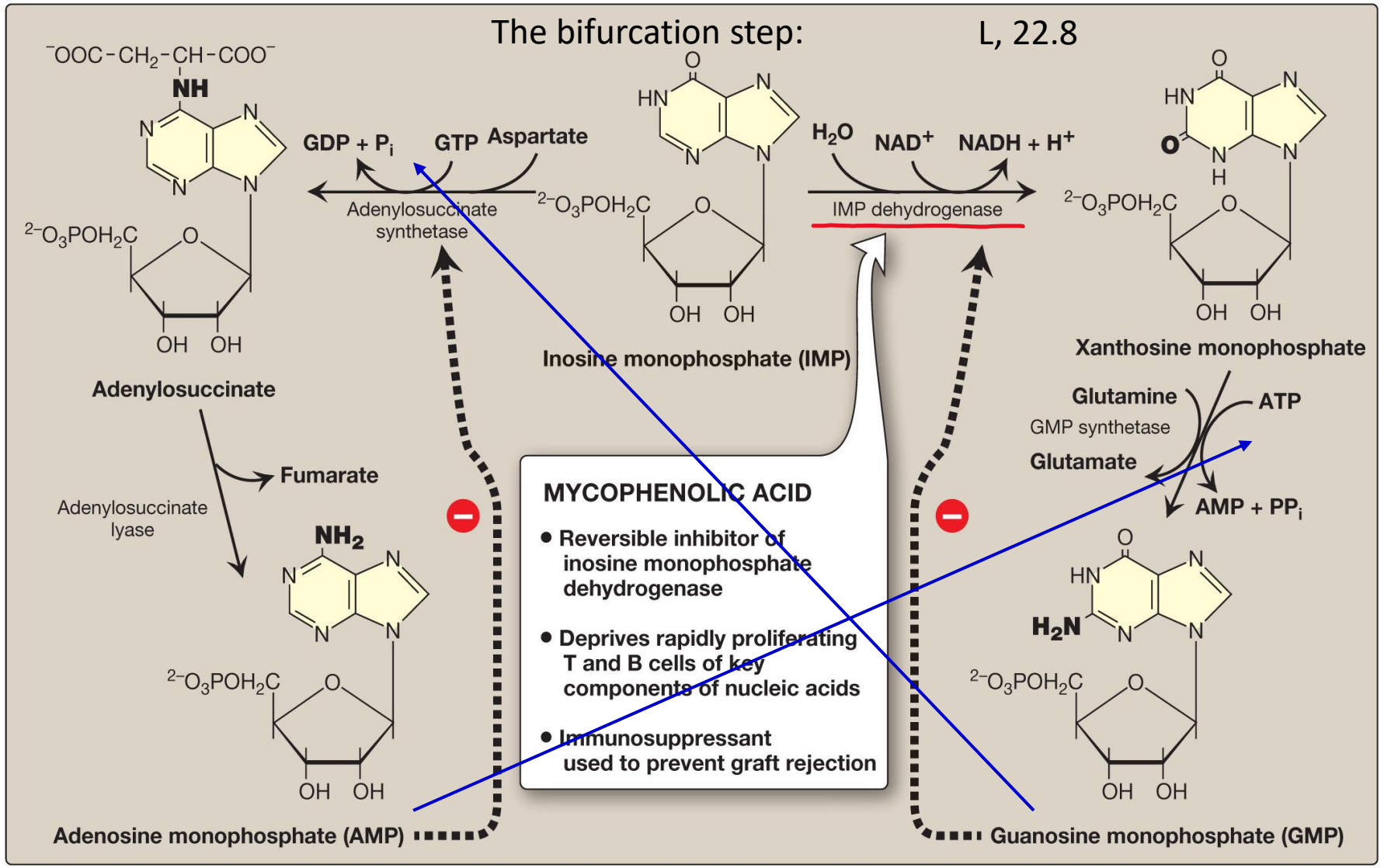
Trimethoprim



L, 22.23

Note: For pyrimidines, Methotrexate (DHR inhibitor, humans), Trimethoprim (DHR inhibitor, bacteria) and 5-FU (Thymidylate inhibitor, more universal) act within the same pathway but have different actions! Know these differences!

Clinical implications of purine de novo synthesis: Mycophenolic Acid (drug)

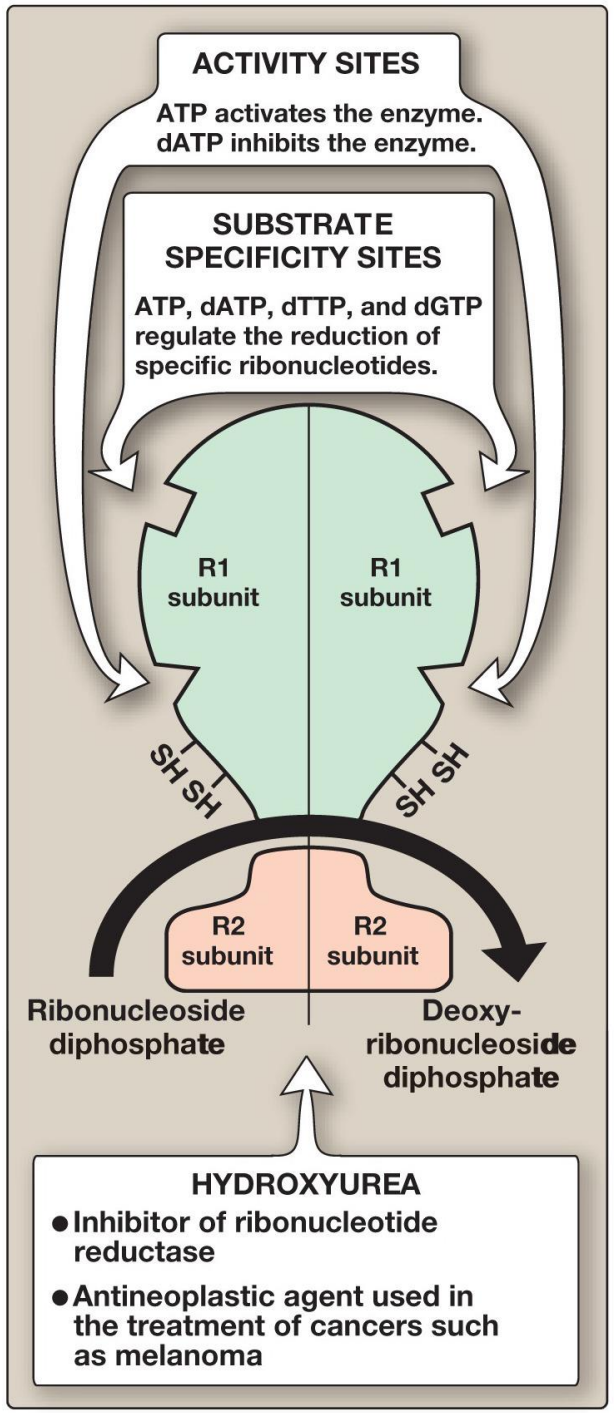


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Mycophenolic acid is a reversible inhibitor of IMP dehydrogenase, the first enzyme used to generate GMP from IMP. GTP is needed to synthesize AMP.

Proliferating T and B lymphocytes are highly susceptible to low levels of GMP.

Consequently, mycophenolic acid is an effective immunosuppressant agent to prevent organ transplant rejection (kidney, heart, and liver) as well as to treat certain immune disorders such as systemic lupus erythematosus and Crohn disease.

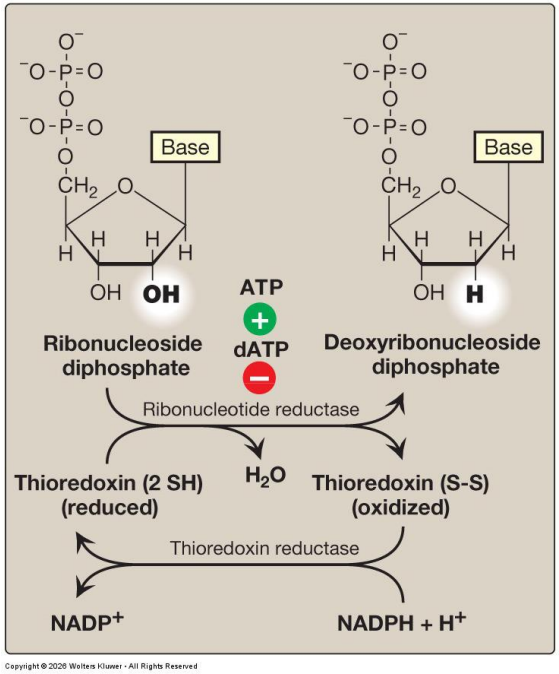


Clinical implications of **deoxyribonucleotide regulation. Hydroxyurea (drug)**

The drug hydroxyurea (hydroxycarbamide) inhibits the ribonucleotide reductase (RR), thereby inhibiting the generation of substrates for DNA synthesis.

The drug is an antineoplastic agent and is used in the treatment of cancers such as melanoma.

Hydroxyurea is also used in the treatment of sickle cell anemia.



L, 22.12

L, 22.13

Drug Biochemistry: Key Points

Sulfonamides: PABA analogs. Block folate synthesis in bacteria, not humans. Prevent bacterial DNA synthesis. **Antibiotics**.

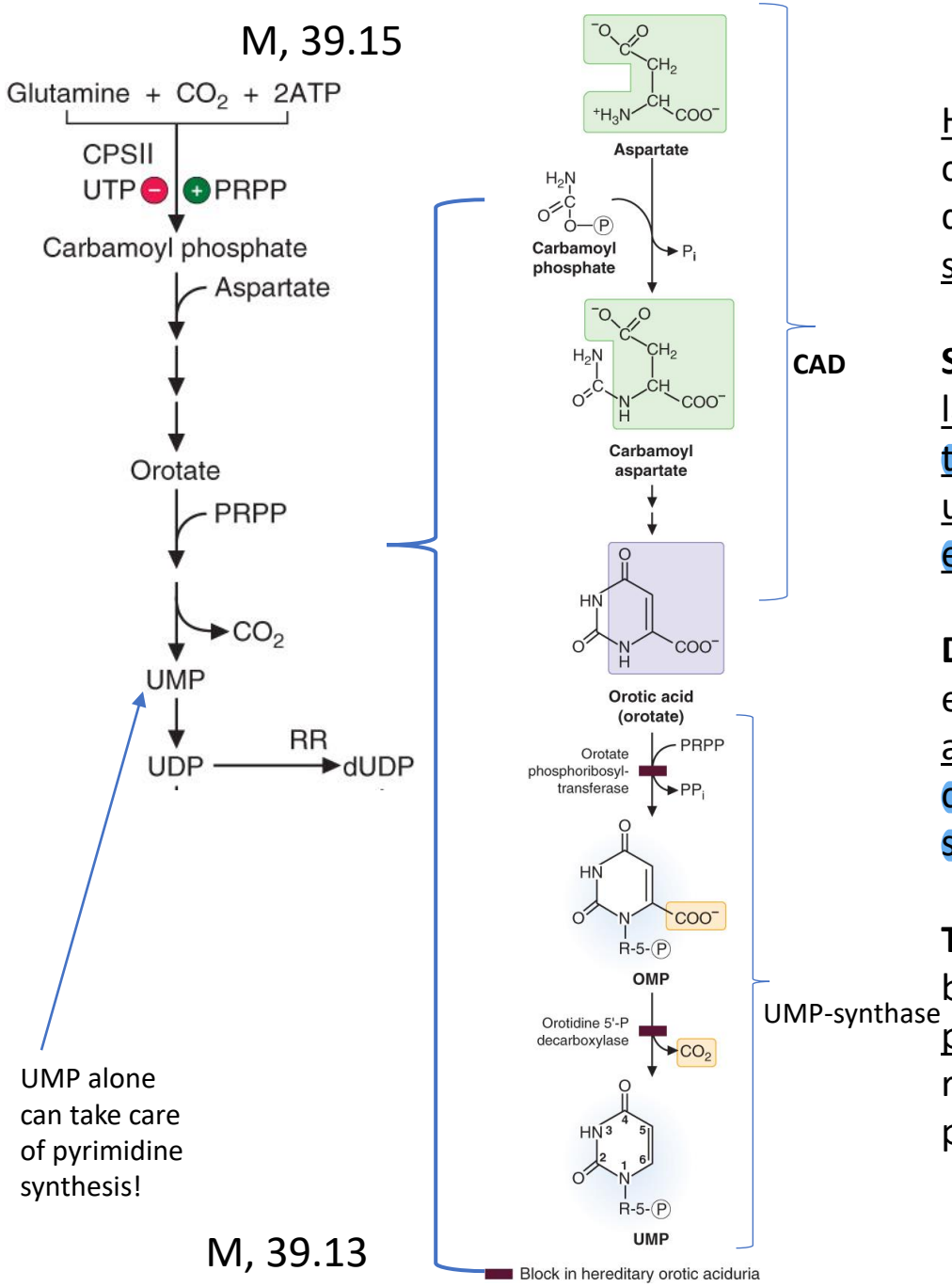
Methotrexate: structural folate analog. Inhibits the dihydrofolate reductase (DHR). High affinity for human DHR. **Anticancer drug**.

Trimethoprim: inhibits the DHR; much higher affinity for the bacterial DHR (~30,000 higher) than for human DHR. **Antibiotic**, especially UTI. Often used with PABA analogs (synergy).

5-Fluorouracil. Suicide inhibition of thymidylate synthase. **Anticancer drug**.

Mycophenolic acid: Reversible inhibitor of IMP dehydrogenase. Proliferating T and B lymphocytes are the most affected. Effective **immunosuppressant agent**.

Hydroxyurea: inhibits the ribonucleotide reductase. Generation of substrates for DNA synthesis is impeded. **Anticancer drug**.



Pyrimidine synthesis, clinical: hereditary orotic aciduria

Hereditary orotic aciduria: orotic acid is excreted in the urine because the enzymes that convert it to UMP, orotate phosphoribosyltransferase and orotidine 5'-phosphate decarboxylase (part of **UMP-synthase**), are defective. Pyrimidines cannot be synthesized.

Symptoms: appear in infancy/early childhood. Include anemia, megaloblastic anemia - large, immature red blood cells, growth failure, cognitive and motor delays, and urinary tract problems (orotic acid crystals can form in the urine, leading to kidney stones, urinary tract infections, and blockage). Other symptoms include diarrhea, vomiting, encephalopathy (brain damage), seizures, heart defects, liver problems, and joint pain.

Diagnosis: hematocrit will show the anemia; peripheral blood smears and bone marrow examination will indicate the megaloblastic anemia. If it does not respond to iron, folic acid, or vitamin B_{12}, orotic aciduria may be suspected. Analysis of the urine (urinary crystals for orotic acid will confirm the suspicions) and analysis for very low UMP synthase activity will establish the diagnosis.

Treatment and Therapy: Uridine, which is converted to UMP, bypasses the metabolic block and provides the body with a source of pyrimidines, as both CTP and dTMP can be produced from UMP. This will eliminate anemia, permit normal development, and reduce orotic acid excretion. The treatment is expected to be lifelong. Many treated patients have led normal lives and live into their late thirties.

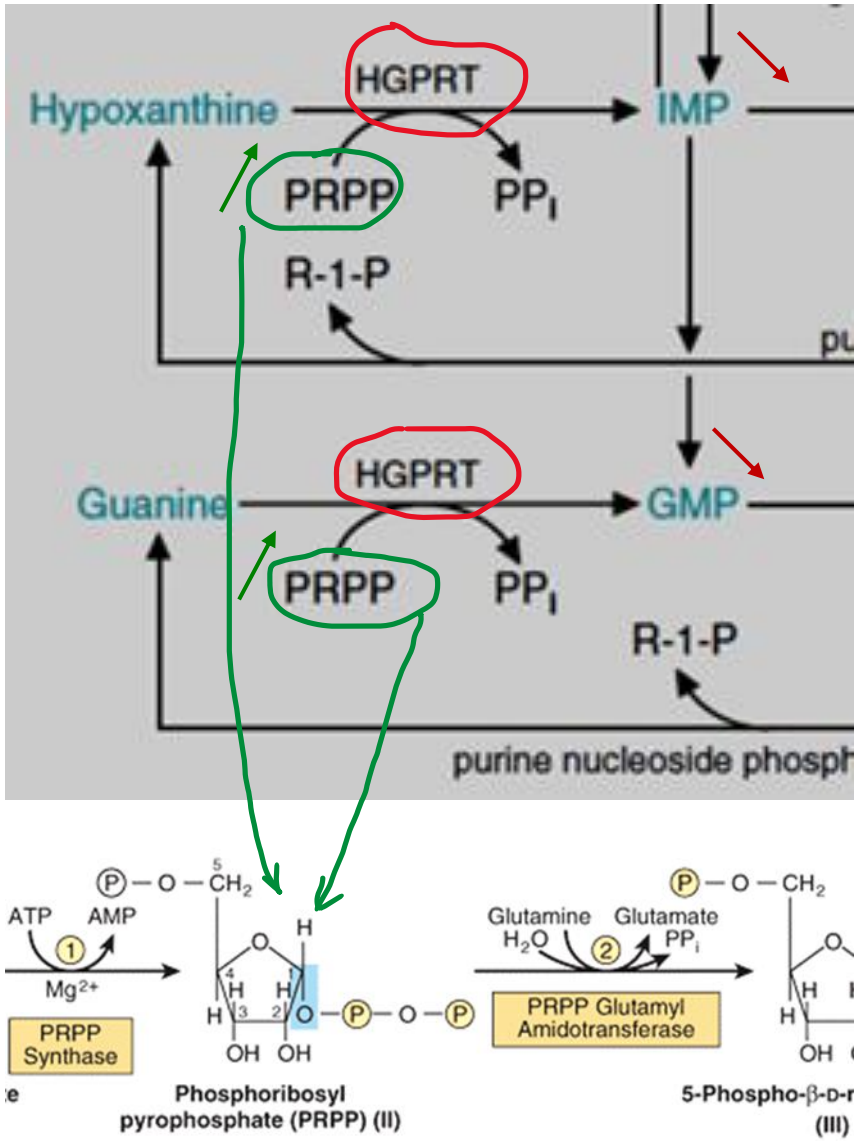
Q: what would be the similarities/differences of manifestations when compared to a deficient CPSII/CAD?

Purine salvage – Clinical Lesch-Nyhan syndrome



L, 22.11

M, 39.10



- **Severe/complete deficiency of HGPRTase activity in the salvage pathway**
  - **X-linked, recessive**
- **~200 known mutations may lead to the loss of HGPRTase, loss of activity, " *K*m mutants," HGPRTase with a short half-life, and so on**
- **HGPRTase catalyzes conversion of hypoxanthine (HX) and guanine (G) to nucleotides. HX and G are not salvaged → increased PRPP and decreased IMP or GMP → intense *de novo* synthesis (with no proper regulation at PRPP – Glutamyl-Amidotransferase, the rate-limiting step).**
- **Symptoms: Hyperuricemia (early gout, urolithiasis - see later) and neurological problems: motor dysfunction, mental retardation, self-mutilation.**

Purine Salvage/Degradation – Clinical

Adenosine Deaminase (ADA) Immunodeficiency

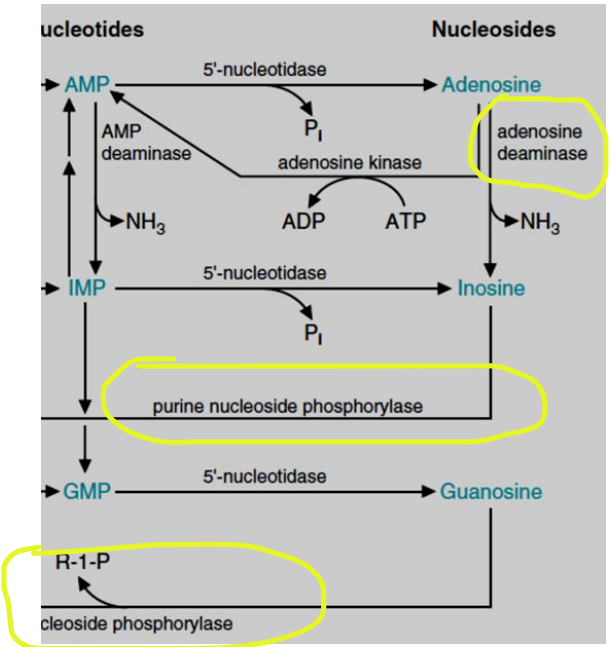
Purine Nucleotide Phosphorylase (PNP) - Immunodeficiency

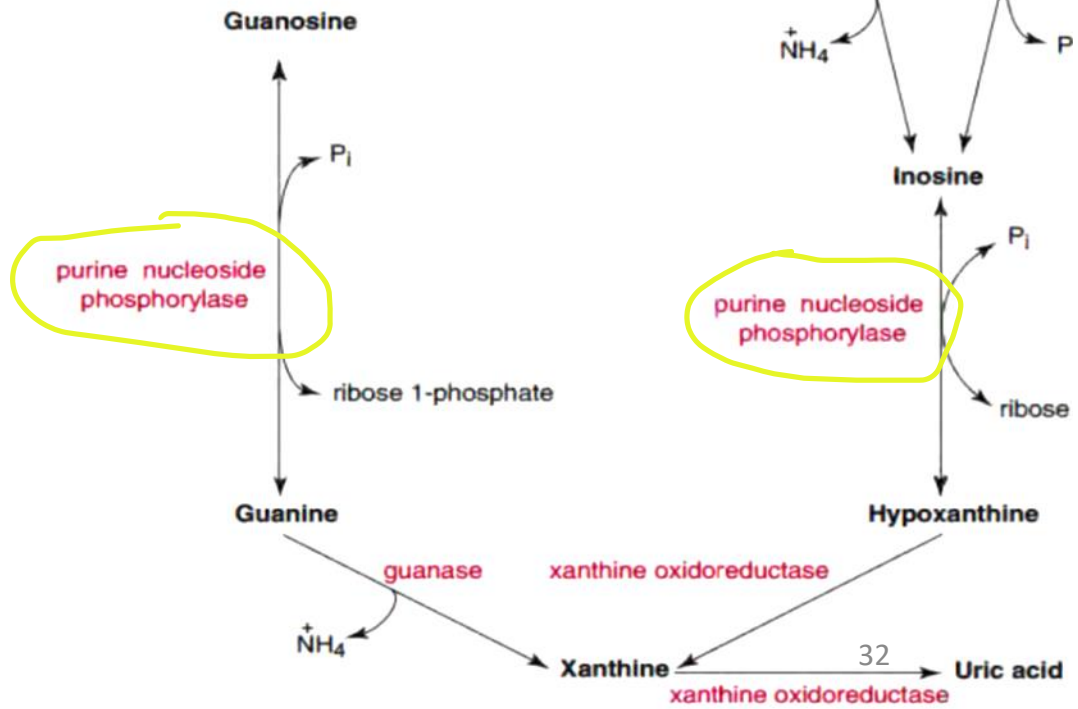
ADA/PNP: involved in salvage and degradation pathways.

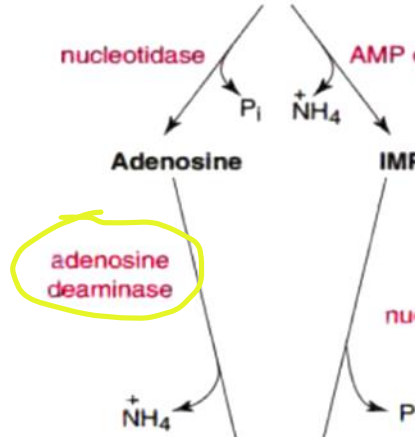
A deficiency in ADA (autosomal recessive) is associated with a severe combined immunodeficiency **(SCID)** involving **both T-cell and B-cell functions**. 15% of SCID

Consequences: Infections, delayed growth and development.

Treatment options: blood transfusions, bone marrow transplantation, enzyme replacement therapy with **ADA-polyethylene glycol conjugate** (ADA-PEG)





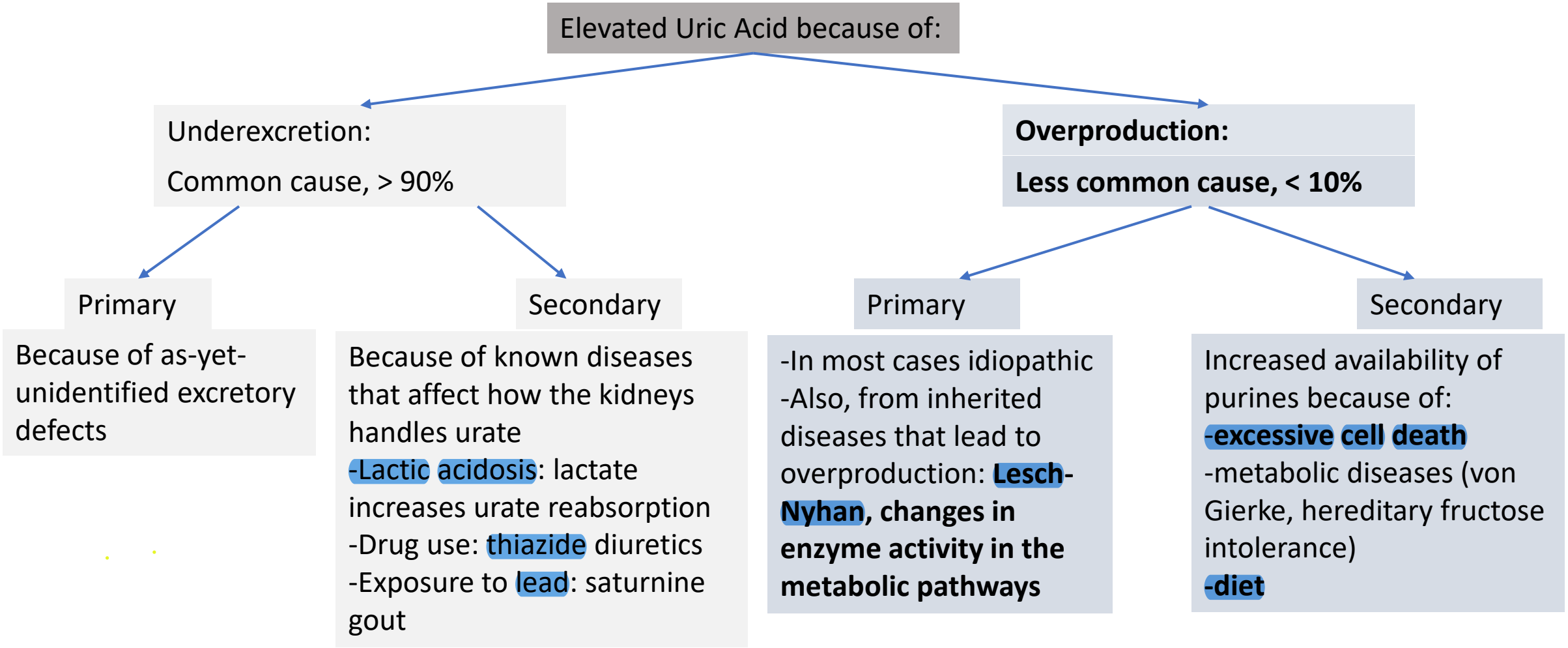


PNP deficiency (autosomal recessive). Associated with an **immunodeficiency** involving **T-cell functions with little or no effect on B-cell function**. 4% of SCID.

Consequences: recurrent infections, and more than half display neurologic complications. Symptoms of the disorder first appear at between 6 months and 4 years of age.

The only **curative treatment** for PNP deficiency is a **hematopoietic stem cell transplantation (HSCT),** which replaces the patient's deficient cells with healthy ones carrying the functional PNP enzyme, effectively correcting the immune system dysfunction caused by this deficiency.

Purine degradation: Hyperuricemia (plasma urate > ~65 mg/L)

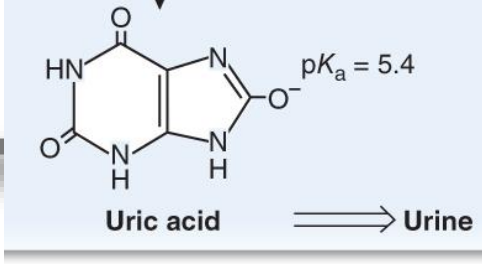


Excessive cell death/Diet

- **Tumor lysis syndrome: Patients undergoing chemo/radio/immuno therapy** – massive cell lysis - increased availability of potassium, phosphate, and nucleic acids into systemic circulation – secondary hyperuricemia
- Similarly, **myeloproliferative disorders.** Cause - high turnover rate: rapid proliferation and destruction of excess blood cells in the bone marrow break down large amounts of nucleic acids - secondary hyperuricemia

- **Diet rich** in meat, seafood (especially shellfish), alcohol consumption, high fructose foods **increases** the risk of hyperuricemia
- A **diet rich** in low-fat diary products **decreases** the risk of hyperuricemia

Uric Acid: Biochemical Properties



- The end-product of purine degradation
- Uric acid is a **weak acid with pKa of 5.5** and a low solubility (~65 mg/L).
- Under physiological conditions (pH 7.4 and 37 °C), uric acid exists as a urate ion.
- In the **crystalline** state, it may exist as **uric acid (mostly in kidney)** and as **monosodium urate crystals (MSU, mostly in joints).** pKa plays an important role in crystallization (see next)!
- Temperature, pH, mechanical stress, cartilage components, and other synovial and serum factors influence development of crystals.
- The crystalline form may **initiate an acute inflammatory reaction** in the synovial joints (gout).
- Crystalline deposits in the urine may result in **urolithiasis and subsequent renal parenchymal damage and kidney stones** over time.
- **Nodular masses** of MSU crystals (tophi) may be deposited in soft tissues (chronic tophaceous gout)


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L, 22.16
Tophaceous gout

Uric acid state explained

- $AH \Leftrightarrow A^- + H^+$
  - Henderson Hasselbalch: pH = pKa + log [A-]/[AH]; pKa = 5.5
- Apply Le Chatelier principle:
  - Neutral pH (blood, synovial fluid): pH = 7.4, >> pKa, ~98 % ionized (A^-); can bind sodium
  - Acidic pH (i.e., < 5.5 i.e., acidic urine), a large portion is protonated (uric acid, unionized, AH); does not bind sodium

- Crystal formation-MSU: neutral pH, A^- may pair with sodium ions to yield monosodium urate (MSU). MSU has low solubility at neutral pH (~65 mg/L); MSU precipitates in joints at higher concentrations (synovial fluid). Why joints? Lower temperature!
- Crystal formation-uric, acid AH - neutral: low pH, kidney. Uric acid has low solubility, ~80 mg/L and may crystalize. The ionized urate (A^-) is more soluble at this pH (~1580 mg/L) so urate does not easily crystalize! Uric acid does!

Uric acid and gout

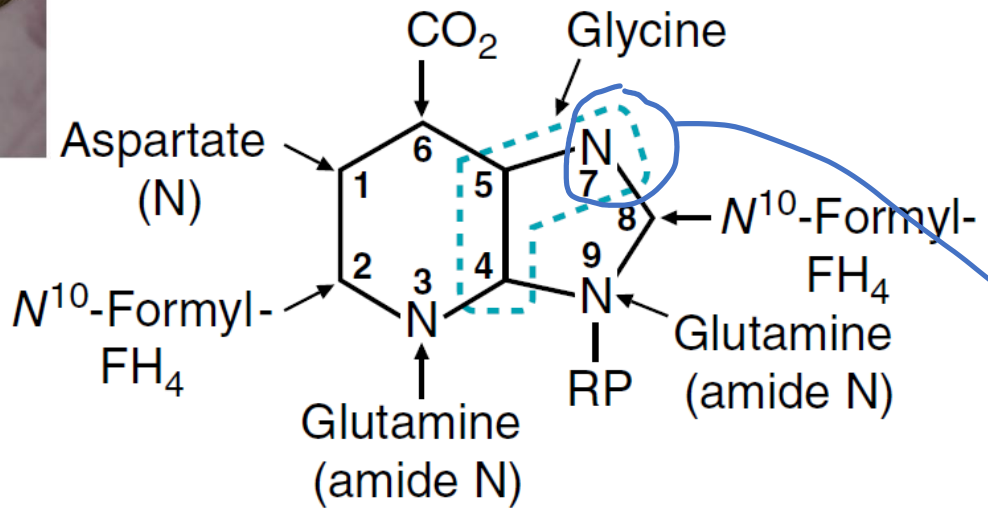
Urate crystal accumulation in joints of the extremities → Triggers infl**a**mmatory response → **Gout; Gout is accompanied by Hyperuricemia, but Hyperuricemia does not necessarily lead to gout**

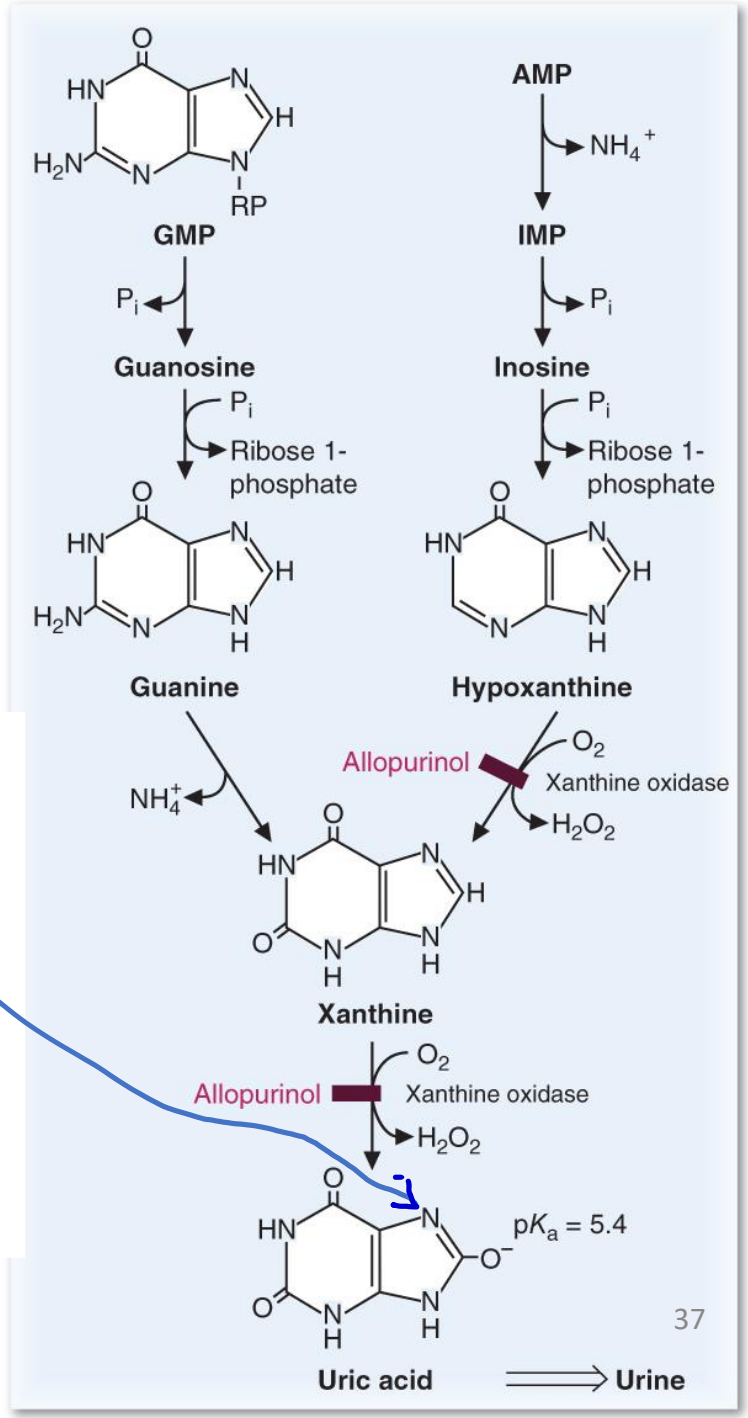
Causes: decreased excretion (renal disease) - 90%, **excessive cell death (i.e., cancer treatment); risk factors (diet); purine overproduction (*de novo*) - 10%**





How to identify if de novo synthesis is the cause?
(look for N7: N^{15} –labeled-glycine)





Purines,Hyperuricemia, Gout

Overproduction of Purines: Possible causes

A) Partial decrease in HGPRT-ase activity: Two fallouts with respect to the crosstalk between *salvage* and *de novo* synthesis pathways:

1) **Decreased salvage of hypoxanthine and guanine** → PRPP is not consumed → **PRPP can enhance PRPP-Glutamyl-amidotransferase** activity → **increased flux through the *de novo* pathway**

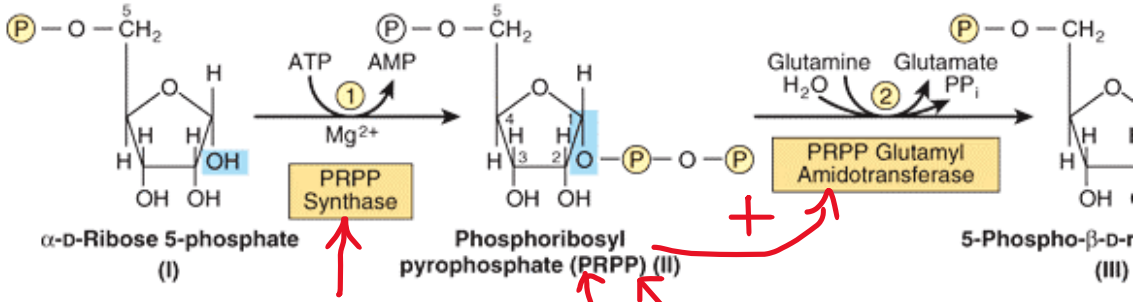
2) **Decreased salvage of hypoxanthine and guanine** → IMP and GMP are not formed → regulation of the **PRPP-Glutamyl-amidotransferase** step by IMP and GMP as negative effectors is compromised → **increased flux through the *de novo* pathway**

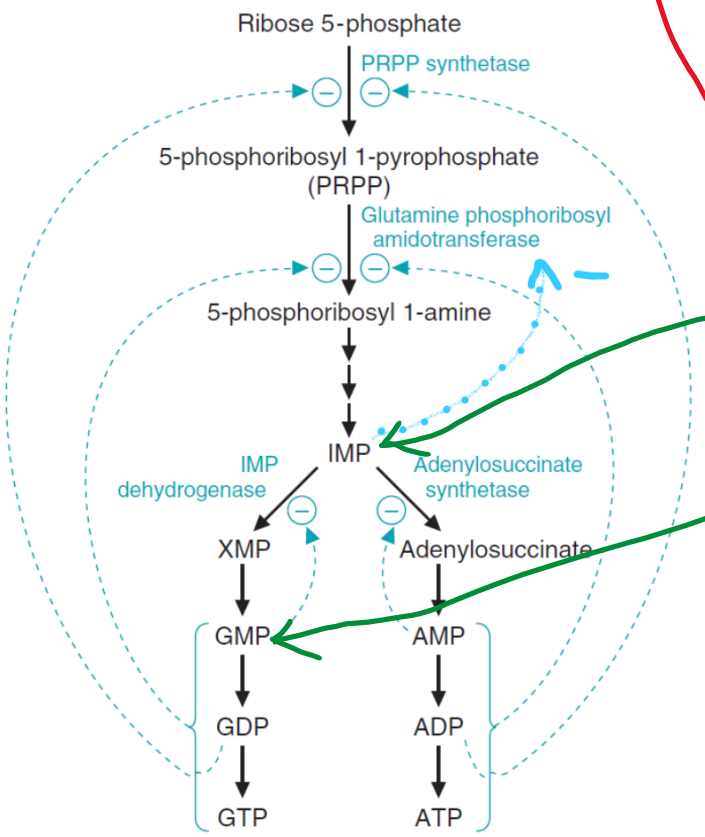
B) PRPP Glutamyl-amidotransferase related:

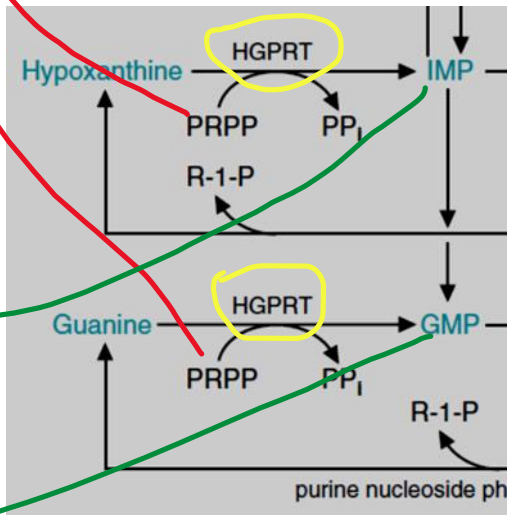
1) Increased PRPP synthase activity → increased intracellular levels of PRPP → **positive effector of PRPP-Glutamyl-amidotransferase** → **increased flux through the *de novo* pathway**

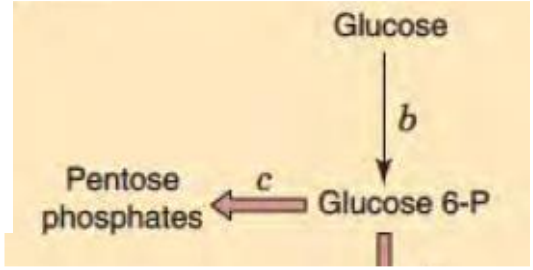
2) G6P-ase deficiency (type I glycogen/von Gierke disease). Glucose 6-phosphate is shunted and passed to the hexose monophosphate path → increased hexose monophosphate shunt activity → **more ribose 5-phosphate → more PRPP.** PRPP is a positive effector of **PRPP-Glutamyl-amidotransferase** → **increased flux through the *de novo* pathway**

All: related to PRPP-Glutamyl-amidotransferase regulation (committed step)!

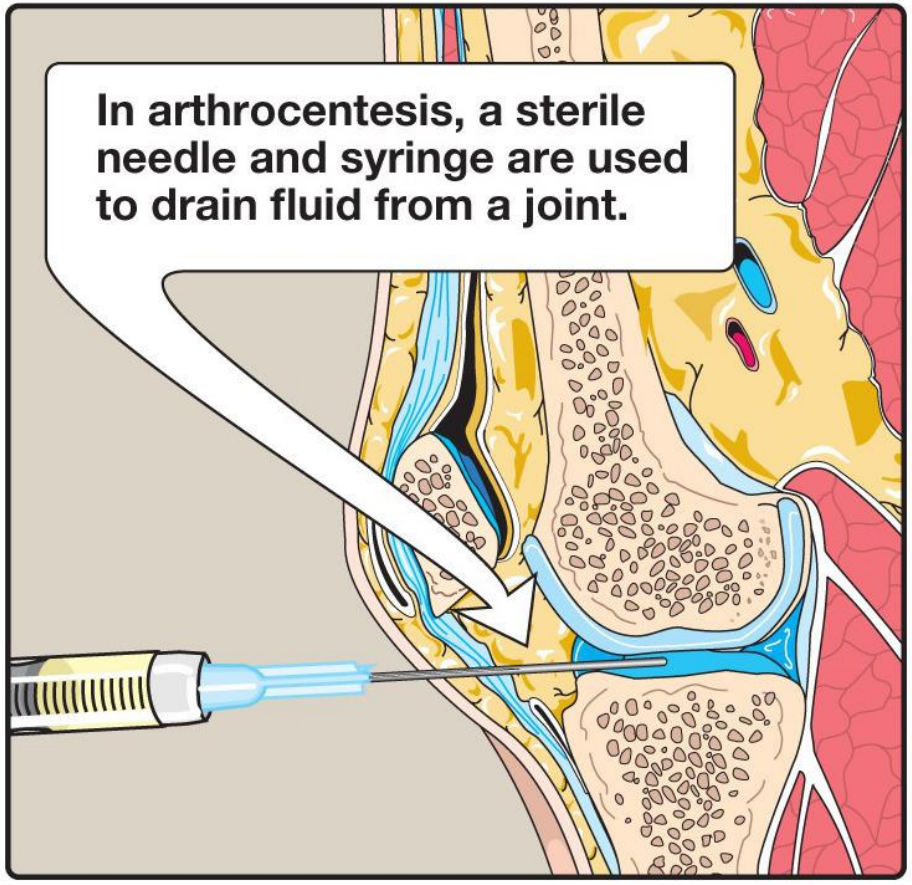








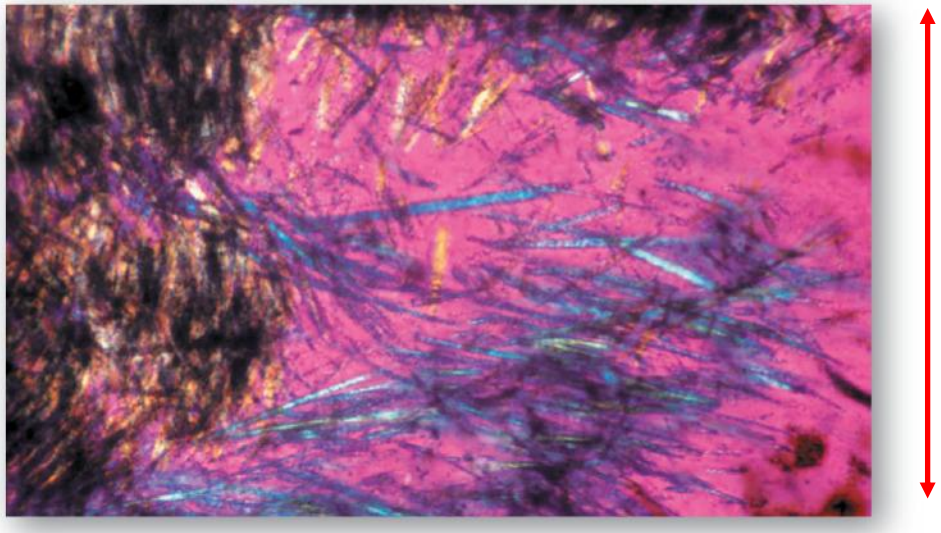
Gout diagnosis: Optics



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L, 22.17

L, 22.18


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Gout can be diagnosed by the presence of **needle/rod shaped, *strongly negatively birefringent* monosodium urate crystals in aspirated synovial fluid examined by *polarized light* microscopy. Crystals parallel to the polarization are yellow; crystals perpendicular to the polarization are blue. This is the gold diagnosis standard for gout.**

Analysis of joint fluid can help to define causes of joint swelling and arthritis, such as infection, gout, and rheumatoid disease.

Gout Treatment

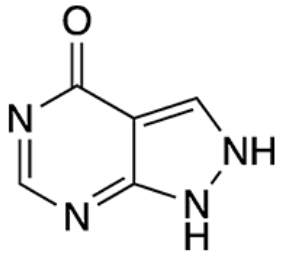
Target inflammation, enhance excretion, **target urate deposit formation (inhibitors, solubilization factors). Urate target : under the solubility limit!**

Allopurinol (purine analog) has a dual role: **competitive inhibitor, and XO substrate**; the metabolite oxipurinol (alloxanthine) binds tightly the XO and inactivates it - **suicide inhibitor**. The concentrations of **hypoxanthine and xanthine increase, urate decreases;** the purine bases are often salvaged to form mononucleotides (HGPRT-ase must be fully functional), and these reactions **consume PRPP** and generate **inhibitors** of PRPP-glutamyl-amidotransferase (i.e., AMP, GMP, acting in the ***de novo synthesis*** pathway, recall regulation).

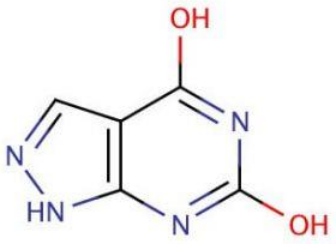
Febuxostat: non-purine analog, inhibitor of XO, non-competitive/mixt inhibition; efficient at lower doses

Solubilization: recombinant PEG-uricase (UA is converted to allantoin, a more **soluble** product)

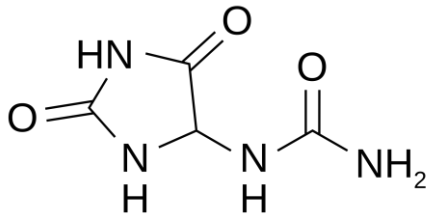
Do not forget diet! Gout = Rich People Disease!



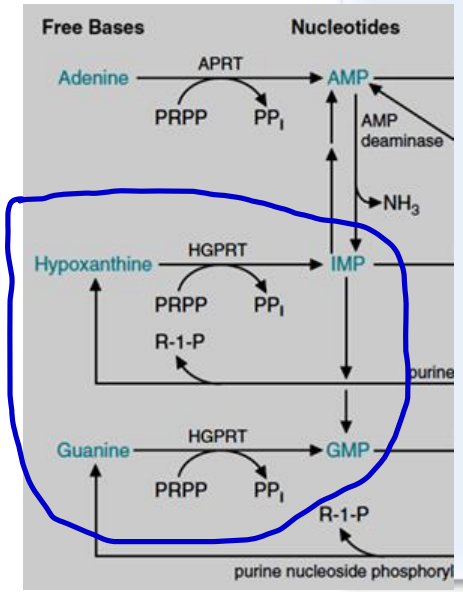
Allopurinol

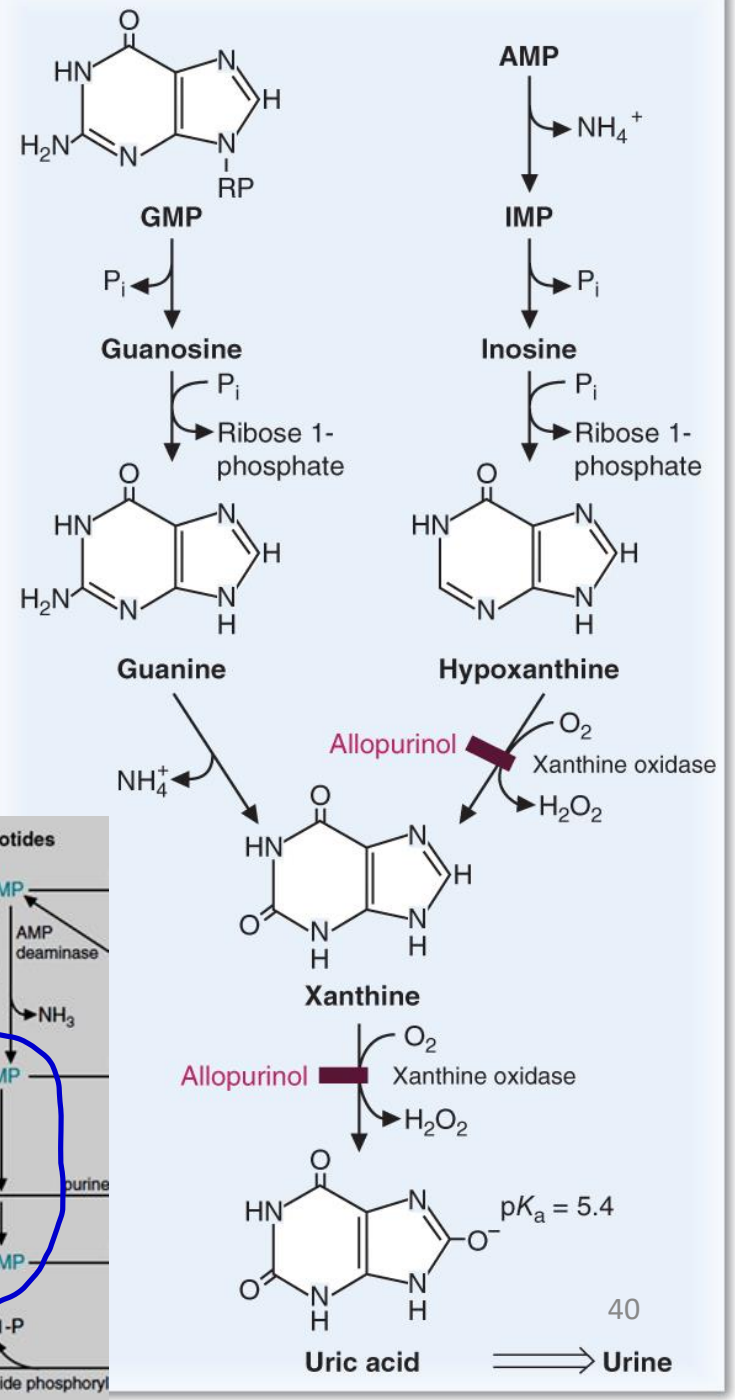


Oxipurinol



Allantoin





Take home lessons

- The purine ring is built step-by-step on the ribose scaffold by addition of atoms and groups from amino acids, CO_2, H_2O; synthesis, salvage, and degradation of purines are strongly integrated
- The pyrimidines are synthesized by attaching the base to the ribose
- Tetrahydrofolate is an important one-carbon donor
- Understanding the pathways led to development of antibiotics and anti-cancer drugs (check the Drug Biochemistry: Key Points slide)
- Genetic defects in metabolically-active enzymes may lead to serious disorders: Lesch-Nyhan, ADA/PNP deficiency; get familiar with causes, symptoms, treatment
- Gout is a result of the inflammatory response to urate accumulation in joints; know very well the causes, symptoms, treatment

References and further reading

- Note: the figures/tables are from ref. 1 (index ”M”) and ref. 2 (Index “L”).

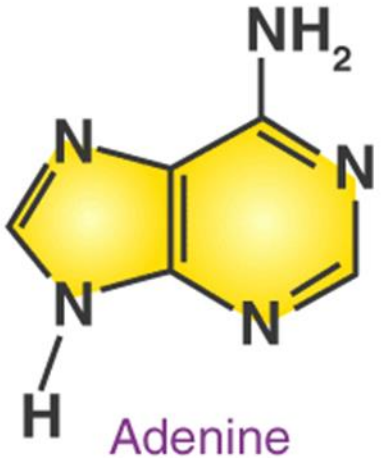
- **1. “Marks’ Basic Medical Biochemistry: A Clinical Approach, 6e”, Chapter 39**

- 1.1 If a review of tetrahydrofolate is needed, check Chapter 38 in ref. 1

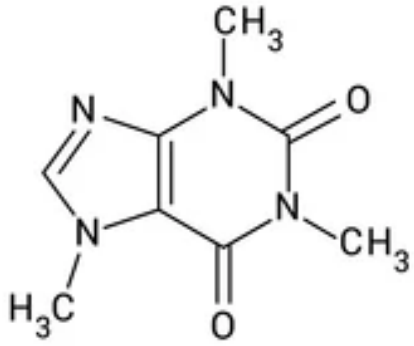
- **2. “Lippincott Illustrated Reviews: Biochemistry”, 8e: Chapter 22**

Thank you!

Your turn: Q&A



Adenine





On a related note: this widely used alkaloid (right) is produced in some plants from adenosine... What is this mystery compound?



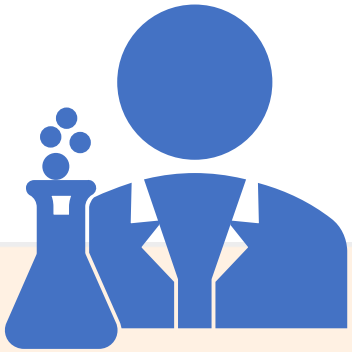
Additional knowledge

A big steak = ~ 2 mMoles nucleotides
This more than doubles for the same amount of shellfish!

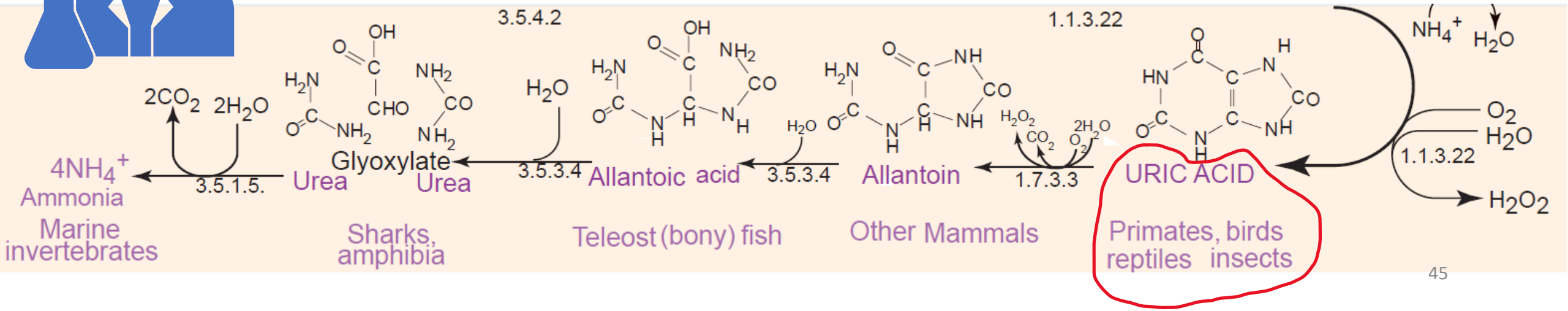


What do you think?

- **Emperor Charles suffered from gout. He was wearing a glove in this Titian's painting. The glove is only on his left hand. Was this a gout recovery phase or a flare?**
- **Why some enzymes (ADA, uricase) used for treatment of purine-related disease are Pegylated?**
- **Humans have uric acid as the final purine degradation product. This may lead to health issues. Was the mother nature wrong? Is there an advantage of having uric acid as the final product of degradation as opposed to a more soluble product?**

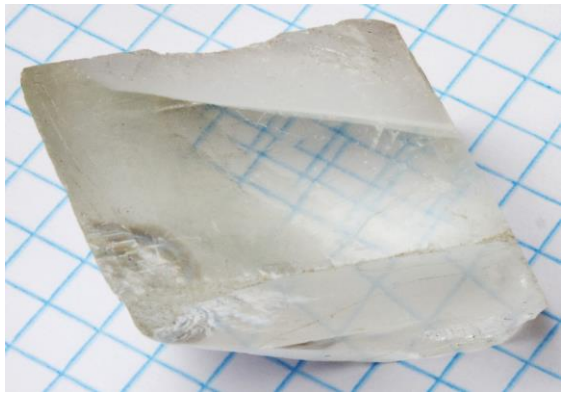


Mammals (other than primates) express urate oxidase (uricase), which cleaves the purine ring, generating more soluble allantoin. Modified recombinant urate oxidase is now used clinically to lower urate levels.



Birefringence

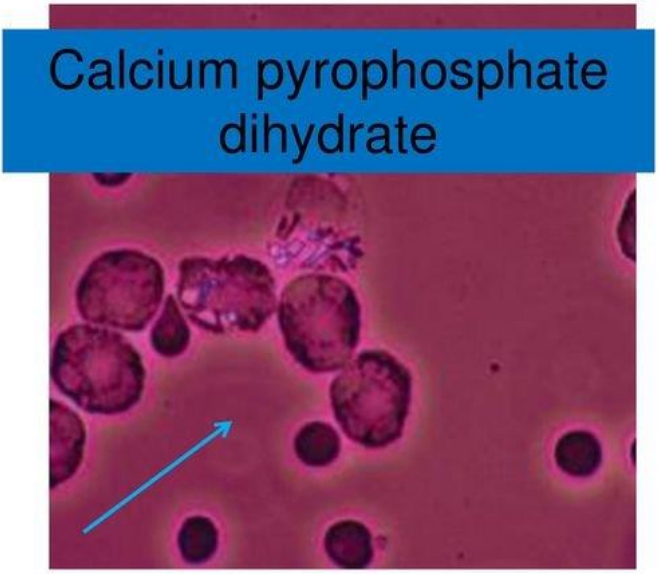
Wikipedia.org



Birefringence, also called **double refraction**, is the optical property of a material having a refractive index n that depends on the polarization and propagation direction of light. These optically anisotropic materials are described as **birefringent** or **birefractive**.

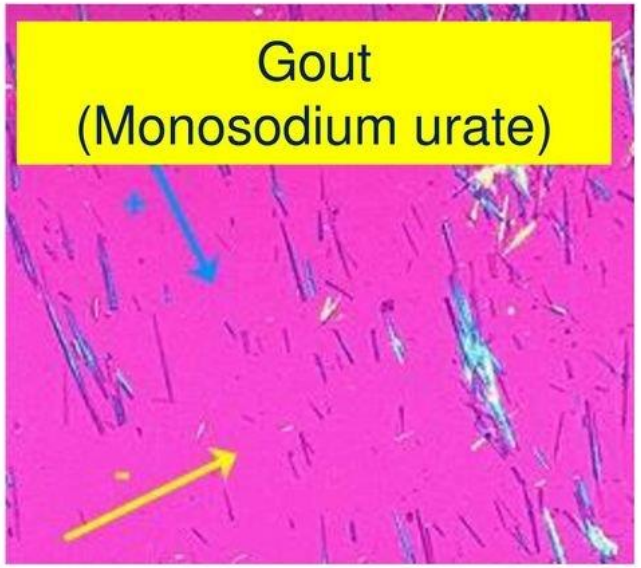
The birefringence is quantified as the maximum difference between refractive indices exhibited by the material. Birefringence is classified as positive when the extraordinary index of refraction n_e is greater than the ordinary index n_o. Negative birefringence means that $\Delta n = n_e - n_o$ is less than zero.

CPP Deposition (Pseudogout)

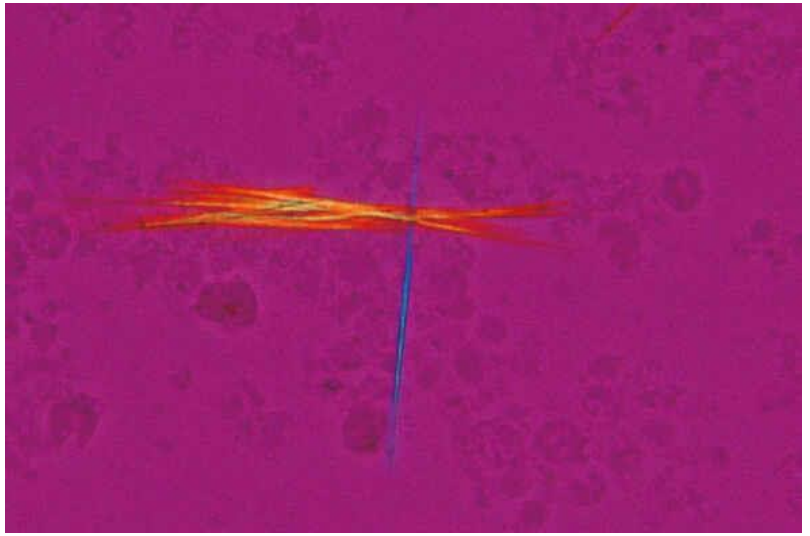


Weakly **positively** birefringent Rhomboid-shaped crystals (Blue when parallel to polarizer)

Gout



Strongly negatively birefringent, needle/rod shaped monosodium urate crystals. Yellow when parallel to polarizer, Blue when perpendicular.



https://www.microscope.healthcare.nikon.com/applications/clinical-research/gout-testing

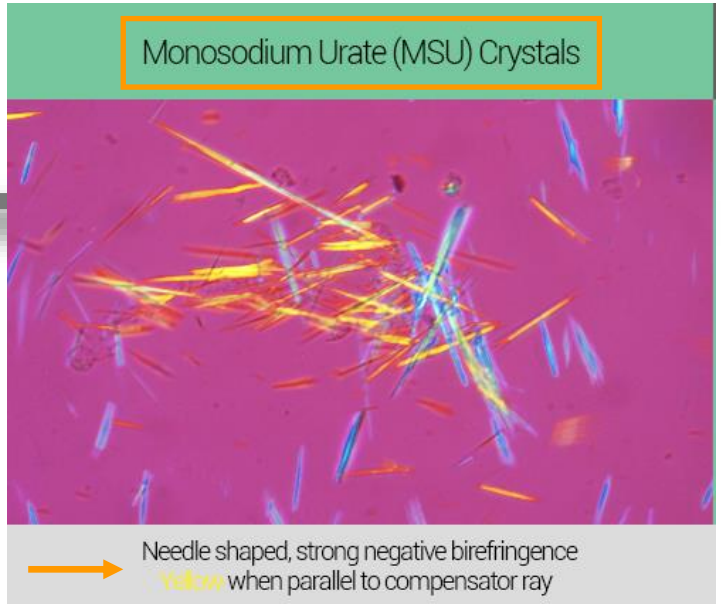
https://x.com/USMLEStepStudy/status/1527248397111021569/photo/2

Gout vs. Pseudogout: Crystals

- **Pseudogout**
- More commonly known to occur in the knee but can be elsewhere.
- May mimic a gout attack (*thus the name*).
- Clinical Correlation: **>50%** of people **80+ years old** will have calcium pyrophosphate crystals found in the joints or surrounding tissues. **25%** of affected individuals will experience symptoms, and **>50%** of the symptomatic group also have concomitant osteoarthritis.

- Pseudogout patients have inherited a predisposing genetic factor.
- Increased occurrence in patients with hemochromatosis.
- Increased occurrence in patients with parathyroid disease.

Gout



Pseudogout

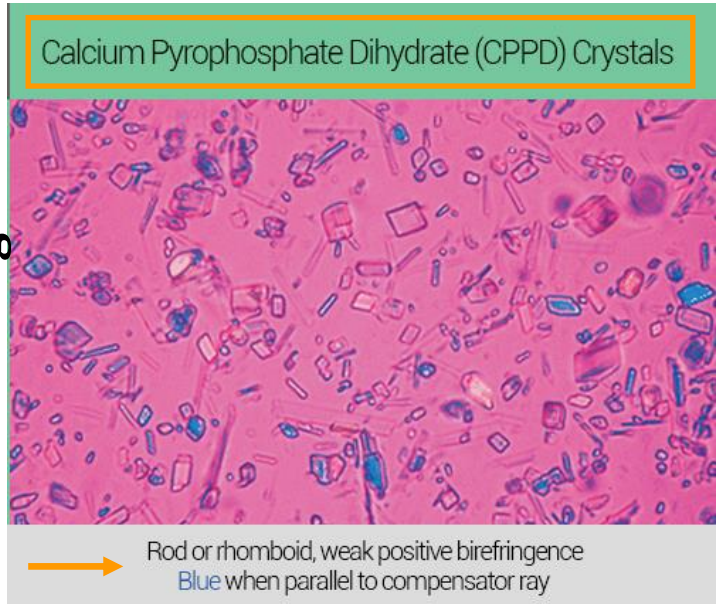


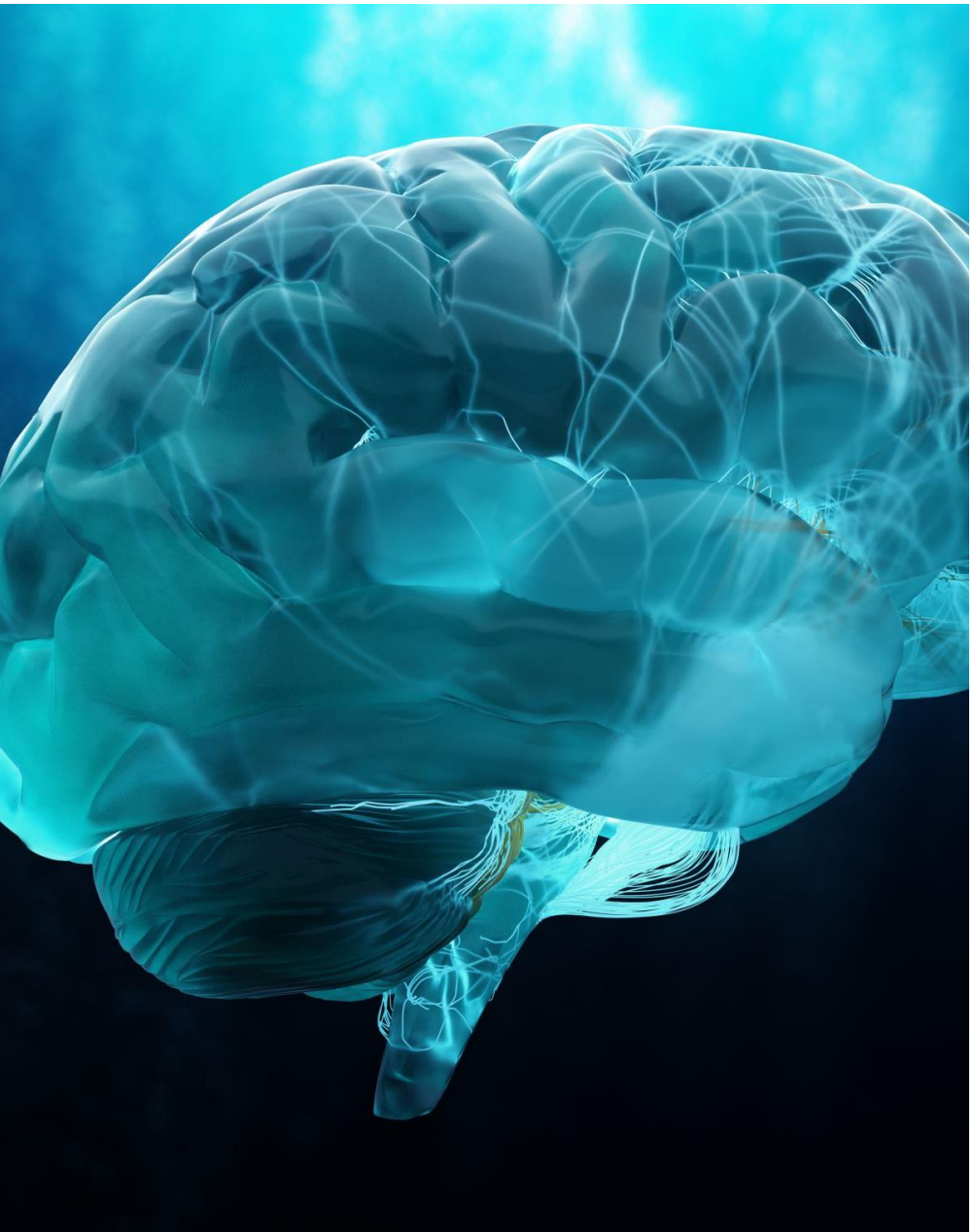
Table: Distinct Crystal Formation Joint/Urinary Tract

Feature	Joint Spaces (Gout)	Urinary Tract (Kidney Stones)
Primary Crystal Formed	Monosodium Urate Monohydrate (MSU)	Pure Uric Acid (Anhydrous or Dihydrate)
Environmental pH	Tight physiological buffer (pH 7.4)	Highly fluctuating, typically acidic (pH < 5.5)
Driving Catalyst	Colder temperatures & high sodium levels	Low fluid volume & hydrogen ion concentration
Microscopic Shape	Long, sharp needles or rods	Rhomboids, diamonds, barrels, or rosettes
Optical Properties	Strongly negative linear birefringence	Strong polychromatic (multicolored) birefringence

Treatment options for Lesch-Nyhan: focused on effects, not cause…

- **There is no successful treatment for the neurological problems.**
- **Reduce the self-injury aspect of the disease.** These treatments have included therapy with levodopa , S-adenosylmethionine, and even injections with botulinum toxin A.
- **Treatment with allopurinol (see later, gout) serves only to decrease the amount of uric acid formed; this relieves some of the problems caused by sodium urate deposits. However,** the *de novo* synthesis of purine nucleotides is not regulated (HGPRTase is not functional, IMP and GMP are not salvaged). This contrasts with patients with gout who will have decreased uric acid levels and decreased *de novo* synthesis in response to allopurinol treatment.
- **The patients usually die from kidney failure, resulting from high sodium urate deposits…**
- **Candidate for gene therapy!**

Lesch-Nyhan syndrome: neurological



- **It is not yet understood why a severe defect in this salvage pathway leads to neurological problems! Here are some potential connections:**

-The normal level of enzyme activity in the brain (frontal lobe, basal ganglia, and cerebellum) is much higher (up to 20 times) than other cells (liver spleen, kidney, RBC).

-It is possible that metabolites of purine degradation are toxic to the developing CNS or that lack of enzyme leads to an imbalance in the concentrations of adenine and guanine nucleotides at critical times during **development**.

-Low concentrations of GTP during development could be the triggering factor in the observed neurological manifestations. GTP is a precursor of tetrahydrobiopterin, a required cofactor in the biosynthesis of neurotransmitters and nitric oxide and is required in other functions such as signal transduction via G proteins and protein synthesis.

- Some data indicate that patients with Lesch-Nyhan disease have **impaired dopamine function.** This could be related to the role of tetrahydrobiopterin in the hydroxylation of tyrosine leading to dopamine synthesis.